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Review

Lifecycle Assessment of Electric Vehicle Lithium-Ion Battery: Current Trends and Future Outlook

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Authors

Dr. Vinotha Krishnasamy

Affiliations

- Sustainable Energy & Power Systems Research Centre, RISE, University of Sharjah, Sharjah, United Arab Emirates

Prof. Mohammad Ali Abdelkareem
Submitting Author

ORCiD
https://orcid.org/0000-0003-3248-9843

Affiliations

- Sustainable Energy & Power Systems Research Centre, RISE, University of Sharjah, Sharjah, United Arab Emirates
- Chemical Engineering Department, Minia University, Elminia, Egypt

Prof. Abdul Ghani Olabi

ORCiD
https://orcid.org/0000-0001-9209-3619

Affiliations

- Sustainable Energy & Power Systems Research Centre, RISE, University of Sharjah, Sharjah, United Arab Emirates

Prof. Jang-Kyo Kim
Corresponding Author

ORCiD
https://orcid.org/0000-0002-5390-8763

Affiliations

- School of Mechanical and Manufacturing Engineering, University of New South Wales, Sydney, NSW 2052, Australia

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Lifecycle Assessment of Electric Vehicle Lithium-Ion Battery: Current Trends and Future Outlook

Vinotha Krishnasamy[1], Mohammad Ali Abdelkareem[1,2*], Abdul Ghani Olabi[1*], Jang-Kyo Kim[3*]

[1]Sustainable Energy & Power Systems Research Centre, RISE, University of Sharjah, Sharjah, United Arab Emirates

[2]Chemical Engineering Department, Minia University, Elminia, Egypt

[3*]School of Mechanical and Manufacturing Engineering, University of New South Wales, Sydney, NSW 2052, Australia

Abstract

The second-life application of lithium-ion batteries after their end of first life in the electric mobility sector is a rapidly growing business opportunity. It requires a deep understanding of the current state of the battery in its first life and accurate prediction of battery performance in its second life. There are several methods for estimating battery health and predicting lifetime. Amongst which the state of health (SOH) estimation and remaining useful life (RUL) prediction are most prevalent methods. They analyse the degradation mechanism and predict battery performance to assess its suitability for different second-life applications. This work overviews recent studies and latest technical developments in SOH estimation and RUL prediction for the second life applications. Data-driven techniques are considered appropriate for second-life battery analysis because they bypass the limitations in conducting experimental studies on used batteries and help reduce the complexity of the experimental methods. Though experimental analysis is widely accepted in the first life of batteries, there are concerns regarding the reliability in the case of used batteries. A survey highlighting the recent trends and new projects in second-life applications is presented. The future requirements for easing the second-life application through implementing digital passports and blockchain are discussed. Overall, the review offers a broad picture of the sequence of events in the LIB lifecycle, starting from the electric mobility sector to second life application, and until it comes to the end of its service life and goes to recycling.

Keywords: Second life batteries (SLBs); State of health (SOH); Data-driven models; Remaining useful life (RUL); Degradation mechanism.

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Acronyms and symbols

AI	Artificial intelligence
ANN	Artificial neutral network
AUKF	Adaptive unscented Kalman filter
BEV	Battery electric vehicle
BiGRU	Bidirectional gated recurrent unit
BiLSTM	Bidirectional long short-term memory
BMS	Battery management system
BPFFNN	Back propagation feed-forward neural network
BPNN	Back propagation neural network
CCA	Calendar aging and cycle aging
CEEMDAN	Complete ensemble empirical mode decomposition with adaptive noise
CEI	Cathode electrolyte interphase
CNN	Convolutional neural network
DEKF	Dual extended Kalman filter
DES	Deep eutectic solvent
DL	Deep learning
DNN	Deep neural network
DOD	Depth of discharge
DP	Digital passport
DRTEM	Dilated residual temporal encoding network with multi-head attention
DWT	Discrete wavelet transform
E-GA	Elitist-genetic algorithm
EMD	Empirical mode decomposition
ENR	Elastic net regression
EOL	End-of-life
ESOI	Energy storage over energy invested
EV	Electric Vehicle
FBG	Fiber Bragg Grating
FL	First life
FLB	First-life battery
FNN	Feedforward neural network
GA	Genetic algorithm
GA-SVR	Genetic algorithm-optimized support vector regression
GHG	Greenhouse gas
GPR	Gaussian process regression

HI	Health indicator
ICA	Incremental capacity analysis
ICEV	Internal combustion engine vehicle
IMF	Intrinsic mode function
LAM	Loss of active material
LCA	Life cycle assessment
LCO	Lithium cobalt oxide
LFP	Lithium iron phosphate
LIB	Lithium-ion battery
LLI	Loss of lithium inventory
LMO	Lithium manganese oxide
LSSVR	Least squares support vector regression
LSTM	Long short-term memory
LTO	Lithium titanium oxide
MAE	Mean absolute error
MAPE	Mean absolute percentage error
MCI	Multi-channel input
ML	Machine learning
MLP	Multi-layer perception
MSE	Mean squared error
NARX	Non-linear autoregressive with exogenous input
NCA	Lithium nickel cobalt aluminum oxide
NMC	Lithium nickel manganese cobalt oxide
NN	Neural network
OEM	Original equipment manufacturer
PINN	Physics-informed neutral network
PHM	Prognostics and health management
PSO	Particle swarm optimization
RFR	Random forest regression
RBAC	Role-based access control
RMSE	Root mean squared error
RNN	Recurrent neural network
RRC	Resistive remaining capacity
RUL	Remaining useful life
SECM	Simplified equivalent circuit model
SEI	Solid electrolyte interphase

SL	Second life
SLB	Second-life battery
SOC	State of charge
SOH	State of health
SVR	Support vector regression
TM	Transition metal
ZEH	Zero-emission house

1. Introduction

The mobility industry is a dominant factor influencing greenhouse gas (GHG) emissions. Sustainability and carbon neutrality initiatives drive the need for the electrification of the transportation sector. Though battery-operated vehicles have been in use for a long period, there has

been an extensive deployment of electric vehicles (EVs) in recent times. In 2022, EVs reached a major landmark, surpassing 10% of global automotive sales. This move is expected to reduce GHG emissions [1]. However, there are other challenges facing the EV industry. The efficiency of EVs needs to be enhanced further to replace fossil fuels completely. There are also issues to be addressed related to the charging infrastructure, drive range, and safety protocols. All these factors are related to the EV battery, which also determines the cost (40%) of the EV [2]. Thus, the EV battery is considered the most prominent part of the EV lifecycle. On average, EVs could have a lifetime of 8 to 10 years, which is close to that of their battery. In the contemporary period, lithium-ion batteries (LIBs) are preferred for EVs. Different cathode materials of LIBs are available in the market, including lithium nickel manganese cobalt oxide (NMC), lithium nickel cobalt aluminum oxide (NCA), lithium iron phosphate (LFP), lithium manganese oxide (LMO), lithium cobalt oxide (LCO), and lithium titanium oxide (LTO). NMCs dominate today's EV market, offering higher energy density and performance. LFPs gain traction in EVs due to their lower cost, longer lifespan, and superior safety profile, making them suitable for vehicles where a slightly lower energy density is acceptable, such as standard-range EVs, buses, and trucks [3].

Efficiency, safety, and reliability are crucial in any battery application. In addition to these factors, the most pressing issue in the EV industry is the safe disposal of EV batteries once they reach end of life (EOL). More EV batteries will be returned from electric mobility in the coming years. By 2030, EV batteries totaling over 17 GWh are anticipated to enter the repurposing stream (**Figure 1)**. Disposing of batteries in landfills poses a serious environmental issue. Battery recycling is a conventional method to recapture most value from the EOL battery. However, there are concerns about the economic feasibility of recycling. There are also concerns regarding the mining of raw materials to produce new batteries. These factors contradict the sustainability and eco-friendly purpose of EVs.

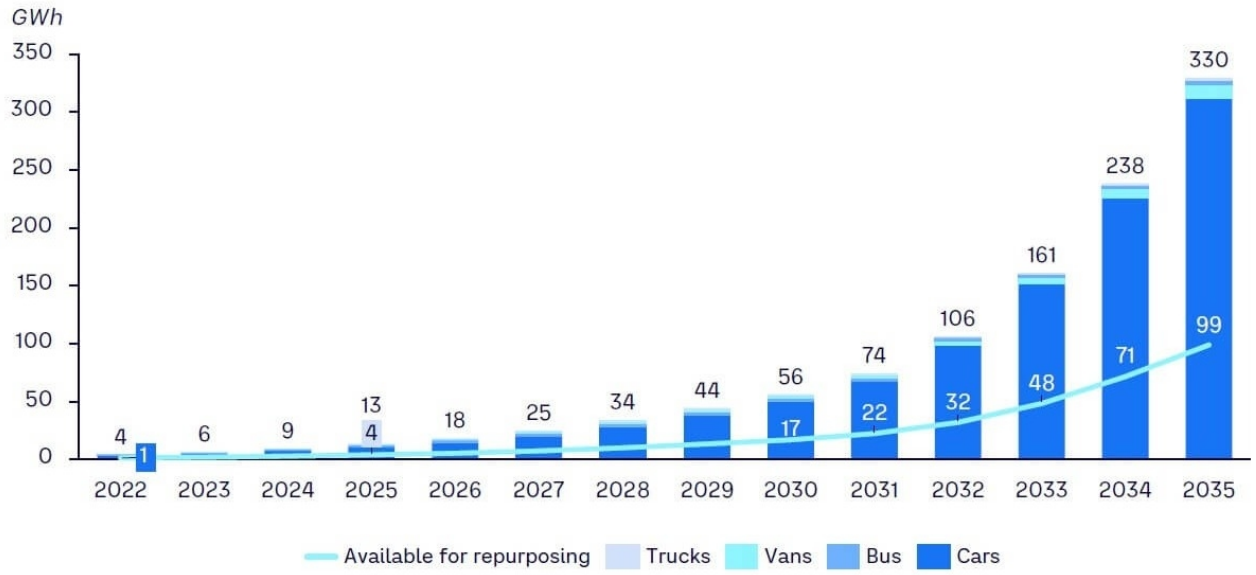


Figure 1: Global returning EV batteries forecast from "Second Life: Maximizing Lifecycle Value of EV Batteries", © Arthur D. Little, 2024. Used with permission [3]

The expansion of the EV market will inevitably result in a higher number of retired EOL batteries. A significant SL for EV batteries can mitigate these effects. The idea is to use high-power EV batteries in less demanding applications [4]. It is predicted that EV batteries reaching 70 to 80 % (EOL in EV application) of their capacity can be transferred to other applications. The lifetime of the battery is extended, and the cost of producing new batteries can be reduced. The industry for second-life battery (SLB) and battery recycling is in its infancy, and more research and analysis are required to develop a competent business model. Major EV manufacturers are running SLB analysis and sorting on a pilot scale. SLBs can be used in low-energy and low-power applications, such as stationary energy storage in microgrids [5], power peak-shaving applications [6], and grid energy arbitrage [7]. SLBs can also be utilized in micromobility applications, like e-scooters and industrial vehicles like forklifts [8].

To mitigate the environmental impact, EV batteries can be remanufactured, repurposed, and recycled based on the battery's state of health (SOH). Each procedure requires fulfilling specific values of power, energy, and life cycle for smooth operation [9]. Under remanufacturing, batteries with more than 80% of the original capacity and power delivery comparable to a new battery can only be considered for reuse in the original application. When the capacity falls below 80%, the EV battery can be repurposed in less demanding applications than EVs. When the battery is no longer suitable for applications in any sector and reaches its EOL, they are recycled to extract critical metals that can be utilized in manufacturing new batteries. The applications of LIBs, based on their remaining capacity, are shown in **Figure 2**.

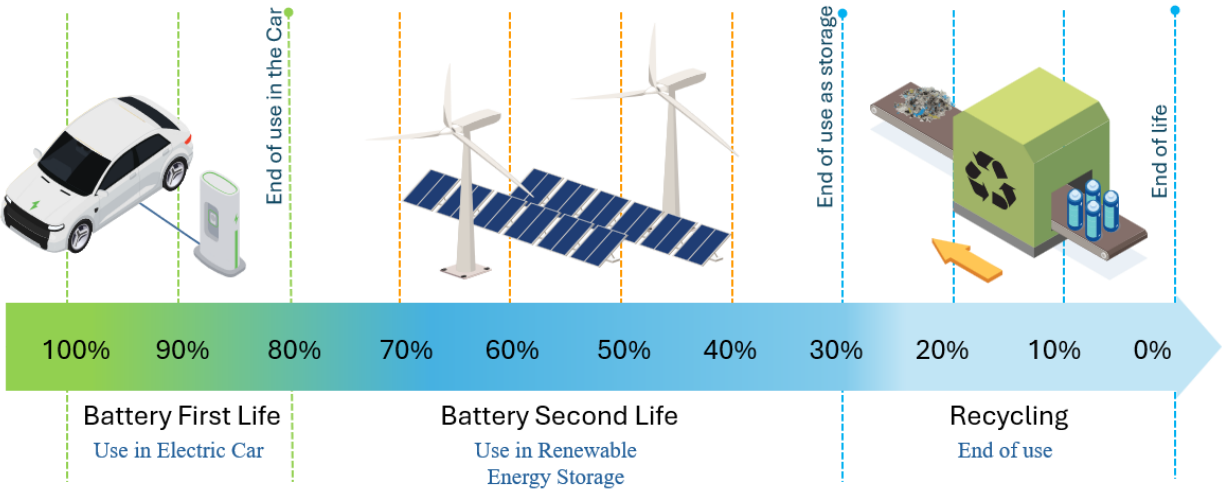


Figure 2: Applications for an electric vehicle battery based on capacity.

An in-depth analysis of the battery condition after the FL is mandatory to decide the significant application in the SL. The battery analysis methods and techniques focus on ensuring efficiency, reliability, and affordability of the battery in an SL application. As per the safety perspective, the thermal runaway is a crucial parameter for setting the SOH lower limit in SLB applications [10]. Thermal runaway is caused by a short circuit or the solid electrolyte interphase (SEI) layer growth. The lower limit for safe operation could be 60% to 40% of SOH [11].

The advantages of reusing a battery in an SL include:

- Reduced expense of the first-life battery (FLB) in EVs
- Possible reduction in environmental impact
- Ensuring better utility value

The focus in this review is placed on the latest trends in SOH assessment and remaining useful life (RUL) prediction in the SL of EV batteries, particularly based on data-driven methods. The degradation mechanism in LIBs is discussed. Key challenges include energy-intensive manufacturing and inefficient recycling, but ongoing research in areas like advanced battery chemistry and improved recycling processes aims to reduce the overall environmental footprint. Advances in SOH estimation methods in the FL are investigated. The latest trends in SOH assessment and RUL forecast are analyzed, and their performances are presented based on error analysis. SLB applications are investigated to understand the applicability and challenges in real-time. The prospects of implementing a digital passport (DP) and blockchain technology for a successful business model are discussed. It includes major EV players and their interest in SL applications.

2. Key life stages and degradation of EV LIBs

2.1. Key life stages of LIBs

The key stages for an EV battery's life cycle include raw material collection and procurement, production process of batteries, including cells, modules, and pack assemblies, integration into the vehicle, the usage phase, and EOL. The EOL stage is further broken down into disassembly, assessment, and either SL applications or recycling. The cradle-to-cradle analysis is required for a holistic understanding of the environmental impact of the EV battery life cycle. It advocates sustainable resource use by incorporating reuse and recycling into the traditional cradle-to-grave assessments. The core indicators are energy consumption, air quality, and GHG emissions. The cradle-to-cradle approach reinvents strategies to improve material recovery and reuse. By improving recycling infrastructure, developing eco-friendly vehicle design, and coordinated effort from other sectors, it is possible to address the environmental impact of electric transportation [12].

2.1.1. Raw material extraction and battery manufacturing

The extraction of raw materials has been a focal point of life cycle assessment (LCA) studies. The environmental impacts connected to the mining and refining of these materials can be substantial. For example, lithium extraction from brine and hard rock mining has significant water use and pollution implications [13]. Cobalt mining occurring in the Democratic Republic of Congo raises ethical concerns related to labor practices and environmental degradation [14]. The transport of raw materials and the final battery products also contributes to the overall carbon footprint.

Battery production procedure itself is a highly energy-demanding process and significantly adds to the overall carbon emission of EVs. A study by Ellingsen et al., [15] quantified the GHG emissions from battery manufacturing, revealing that the production of LIBs can account for up to 50% of the total lifecycle emissions of an EV. This finding underscores the importance of improving battery manufacturing processes and exploring alternative materials to mitigate environmental impacts.

The first cradle-to-gate emission analysis of a high-volume battery manufactured for a commercial battery electric vehicle (BEV) estimates a 39% hike in the cradle-to-gate GHG emissions of the Focus BEV as opposed to the Focus internal combustion engine vehicle (ICEV) [16]. By reducing the uncertainties in the assessment of BEV battery production, this analysis aids in identifying possibilities to curtail emissions and claims that there is a scope to reduce the GHG emissions over the full life cycle of the BEV. The high-level usage of clean energy in the production process is pivotal to mitigating these adverse effects. The entire industrial chain should emphasize eco-friendly progression [17].

2.1.2. Usage phase and energy source

The usage phase of EVs is crucial for understanding their overall positive environmental impacts. The lifespan of the battery during the vehicle's use is impacted by factors such as charging and discharging rates, temperature, and overall usage. A longer cycle life and slower degradation rate lead to a lower environmental impact per unit of energy used. The emissions associated with the operation

of EVs are mainly contingent to electricity generation sources. Studies have demonstrated that in regions where electricity is generated from renewable sources, EVs can substantially lower GHG emissions relative to ICEVs [18] . Conversely, in regions highly dependent on coal or fossil fuels, the emission benefits of EVs diminish [19].

The role of the electricity grid in the LCA of EVs has been highlighted by several studies. For instance, [20] emphasized the importance of considering grid mix when evaluating the environmental benefits of EVs, suggesting that a transition to renewable energy sources is essential for maximizing the sustainability of EVs.

The use-phase energy consumption of EVs also depends on the driving pattern and use of climate control. Aggressive driving is seen to hike energy utilization by 47%, while climate control can bring about an increase of 24 to 60%. Studies suggest three factors for the sustainability of EVs: adoption of advanced battery technology, an eco-driving attitude and a sustainable electricity mix [21]. EV owners benefit from demand response programs, as they provide an incentive for adjusting their charging behavior to comply with grid conditions [22]. This approach facilitates the integration of high clean energy generation with the charging periods of EVs [23]. Battery swapping stations integrated with renewable energy charging can reduce carbon emissions effectively [24]. The vehicle-to-grid (V2G) technology, combined with smart charging strategies with good communication and timely planning, can greatly benefit the grid services using EVs [25]. When EVs are powered using renewable energy sources the environmental impact is substantially reduced [26].

2.1.3. End of Life (EOL) Management

The EOL phase of LIBs presents both challenges and opportunities. They can be either reused in an SL application, such as grid energy storage, or recycled. Cascading use of EV batteries through a hierarchy of SL applications will mitigate life cycle impacts and leverage critical metal use. The energy storage over energy invested (ESOI) metric helps to compare the competence of energy storage technologies. In the case of EVs, the ESOI metric will be better if the battery is used in less critical SL applications. Well-heeled SL applications can potentially offset the cost of recycling by contributing to the value stream.

Recycling is crucial to recover valuable materials and minimize waste, but current recycling processes can be complex and inefficient, though improvements are being made. The disposal and recycling of batteries can result in significant environmental impacts if not managed properly. Current practices often involve landfilling or incineration, which can lead to leaching of toxic substances into the environment [27]. However, advancements in recycling technologies have the potential to reclaim critical materials and lower the environmental footprint of battery disposal.

Several studies have explored the potential of closed-loop recycling systems for LIBs. Recycling could reclaim as much as 95% of critical metals, considerably cutting the demand for virgin materials and mitigating the environmental impacts of raw material extraction [25]. The integration of recycling

into the LCA of EVs is essential for a holistic understanding of their environmental impacts. Studies show that processing mixed-stream LIBs into battery-grade materials minimizes the negative environmental impacts by at least 58%. When considering the recycling process of LIBs, the electricity requirement is the major reason. Using alternative electricity sources can reduce GHG emissions by up to five times [28]. The recycling industry faces many challenges in optimizing the procedure, as different physical configurations, cell chemistries, and cell types are used by different manufacturers. The automation of disassembly is a challenge [29], while the choice of recycling technique is another issue, as they come with pros and cons. Pyrometallurgy and hydrometallurgy are widely used recycling technologies that are currently adopted by industries. The pyrometallurgical process refers to a two-step high-temperature smelting process[30], whereas the hydrometallurgy process involves the recovery of materials utilizing aqueous chemistry [31]. Direct recycling focuses on cathode recovery by targeted healing [32]. Hydrometallurgy is considered better than the other techniques as it involves minimal energy use, excellent metal extraction efficiency, and high-grade purity. However, some of its drawbacks include the consumption of high chemical reagents and the long process duration[33]. Thus, recent research is directed toward using safe pretreatment methods, such as green and reusable deep eutectic solvents (DESs) for leaching, and the adoption of simple extraction techniques for retrieving valuable metals. It is important to develop better recycling infrastructure globally.

2.2. Degradation of LIBs

Since LIBs are most preferred in EVs, understanding the degradation mechanism of LIBs in EVs is important to estimate the future application of the battery or to consider recycling. LIB degradation is a complex process. The underlying mechanism, the interconnections, the triggers, and possible outcomes should be predicted by the different analysis techniques. Deciphering battery degradation aids in the efficient handling of the operational lifetime of LIBs. Accurate prediction of EOL is more beneficial in SLBs than in FLBs [34]. Battery degradation occurs due to the aging of components, and the battery aging phenomena can be classified into calendar aging and cycle aging [35], [36]. Calendar aging occurs when the battery is idle, i.e., not in operation, and cycle aging is associated with the charge/discharge cycles of the battery when it is being used.

2.2.1. Calendar aging: Calendar aging of a battery depends on the temperature and the state of charge (SOC) at different periods [37]. Calendar ageing is important in EVs as there are more idle periods than the operation time. A passivation layer is formed on the anode surface, i.e., SEI layer, due to electrolyte decomposition, leading to cyclable lithium consumption as well as dissolution of transition metal (TM) ions from the cathode [38], [39], [40], [41]. The details of SEI formation are discussed in Section 2.3. Calendar ageing is influenced by high ambient temperature [36] and high SOC storage conditions.

2.2.2. Cycle aging: The battery aging mechanism, when cycle aging is included, is dependent on many factors, and the mechanism differs for different LIB chemistries [42]. Cycle ageing is caused by mechanical stresses occurring in the electrode active material, additives, or lithium plating [43], [44]. The estimation of SOC is complex, as it varies nonlinearly with changes in temperature and applied current [45].

LIB aging effects include the changes in electrode structure, electrolyte composition, and physicochemical properties. Hence, accurate estimation of SOC must include the effects of calendar and cycle aging. Battery aging is influenced most by temperature, followed by voltage, current, and depth of discharge (DOD). Managing SEI growth and avoiding lithium plating are key to improving battery life. Operation at high temperatures improves the SEI layer but accelerates other aging processes, so that there is a trade-off [46]. Aging causes capacity fade and power reduction via an increase in internal resistance in the battery. The material and structural changes in LIB components after degradation, along with their contributions to battery performance in the form of capacity and power drops, are presented in **Figure 3.** The details of degradation mechanisms of battery components by electrochemical cycle aging are discussed in the following section.

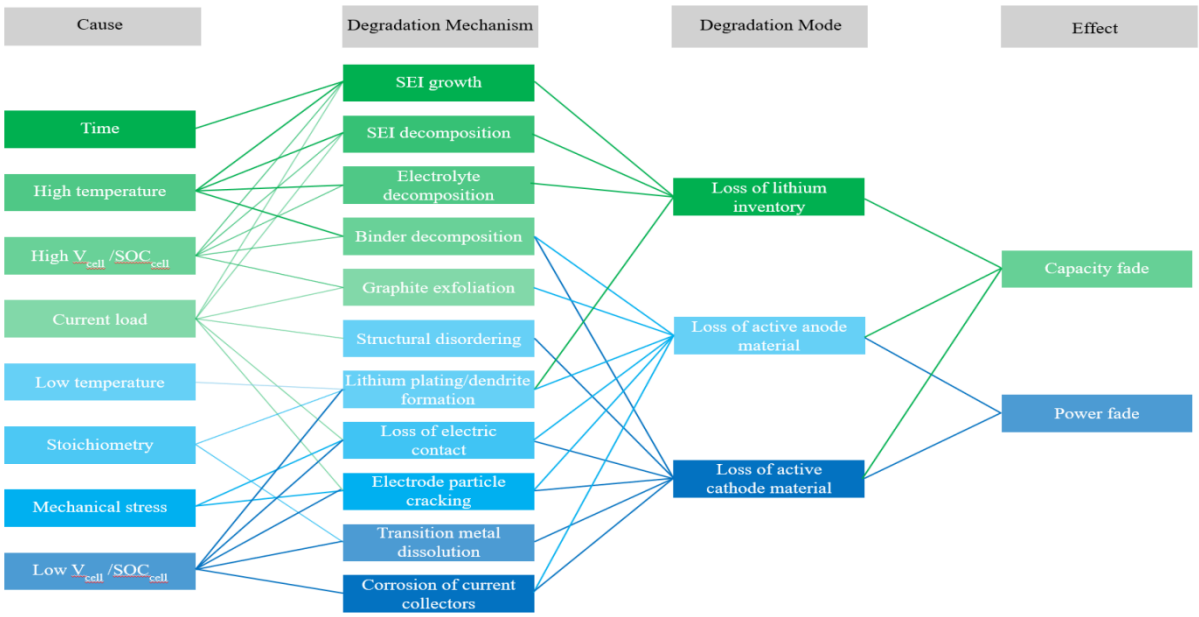


Figure 3: Degradation mechanisms in LIBs. Adapted from [44].

2.3. Factors affecting battery degradation mechanisms

Degradation of a battery refers to a reduction in the overall energy storage capacity, in which the capacity to store and supply energy declines. This degradation negatively affects the operational lifespans of energy storage systems. The degradation mechanisms in intercalation-type LIBs that are commonly used in EVs are discussed in a recent review [47]. The predominant cause of degradation is charge/discharge cycles in their operational lifetime [48], which alter the physical and chemical nature of battery cells. Under adverse conditions, batteries cannot guarantee efficient, sustainable performance. Extreme temperature operation, overcharge or deep charge, SOC, DOD, and calendar aging all accelerate the degradation process, depending on the individual usage scenario [49], [50], [51]. The degradation phenomena taking place in the battery components are illustrated in **Figure 4,** and the details of the mechanisms are discussed in the following.

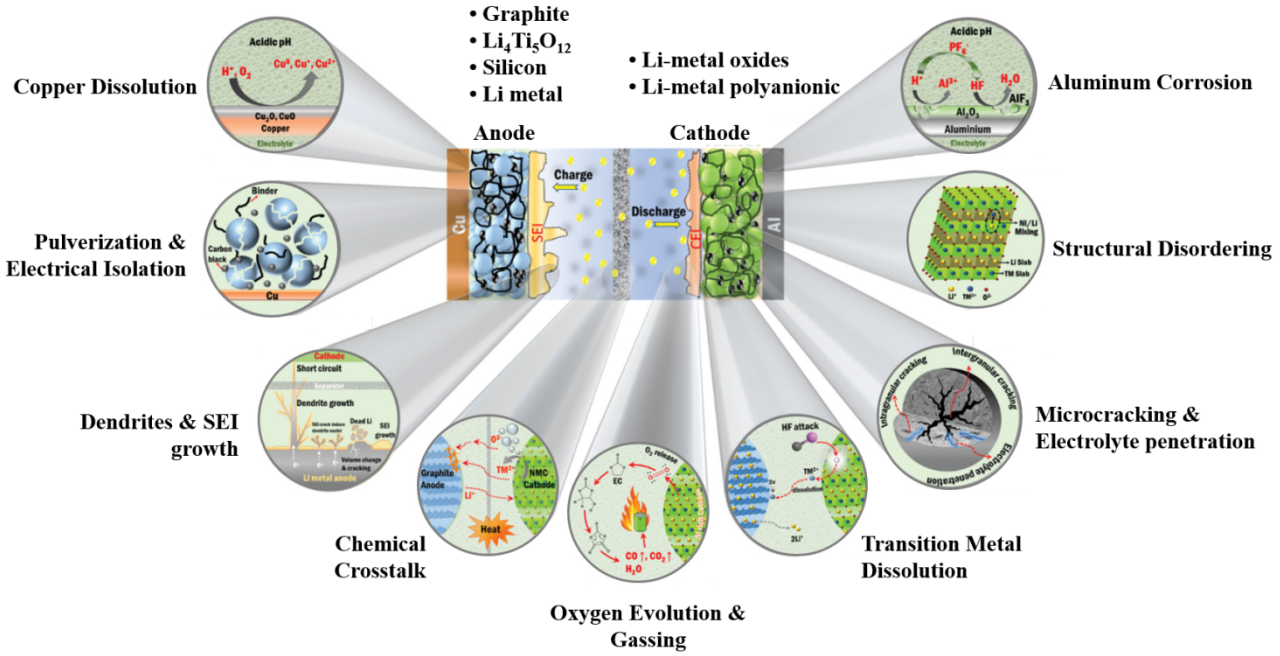


Figure 4: Degradation mechanisms in LIB. Reproduced from [52].

2.3.1. Degradation mechanisms in the anode

The negative electrode in LIBs mainly comprises either graphite, silicon, titanate, lithium metal, or various composite materials [53]. Most batteries have graphite-based anodes because of the high electronic/ionic conductivity, suitable thermal and mechanical properties. The carbon-based anodes are very affordable and easily available in terms of raw material requirements [54]. The battery anode is prone to degradation due to SEI formation, lithium plating and stress generation.

- *Solid electrolyte interphase (SEI):* The SEI forms on the graphite anode during the first charge/discharge cycle. The thermodynamic instability of the graphite anode when in contact with the electrolyte leads to a reductive electrolyte breakdown, creating a thin passivation layer that allows lithium ions to pass but blocks electrons. Lithium ions are consumed in the process, aiding the

formation of SEI. It is useful in preventing direct connection of the electrode with the electrolyte. However, the expansion of the SEI layer due to high temperature, high SOC, and mechanical stress in the electrode due to volume expansion subsequently leads to loss of active material (LAM) and loss of lithium inventory (LLI) [55], [56]. Electrolyte composition and electrode balance are critical for the expansion of SEI.

• *Lithium Plating:* The uneven deposition of lithium ions in the anode generates metallic lithium during charging, contributing to the battery degradation [57]. Low temperature and rapid charging conditions elevate lithium plating. LLI induced by lithium plating leads to the formation of lithium dendrites, which cause safety hazards and performance issues like internal short circuits, fires, and battery failure, by piercing the separator and bridging the anode and cathode [58].

• *Mechanical stress:* The generation of mechanical stresses affects the structural integrity of the anode, which leads to severe deformation or cracking (**Figure 5**). Eventually, the energy storage capacity of the battery deteriorates [59]. The stress-induced dendrite formation also poses a serious safety hazard as it may lead to thermal runaway, overheating, or even the worst-case scenario of explosion [60], [61], [62], [63]. The stresses generated due to volume shift cause surface layer cracking (LLI and LAM), anode structural damage (LAM), and contact loss with the composite electrode [64].

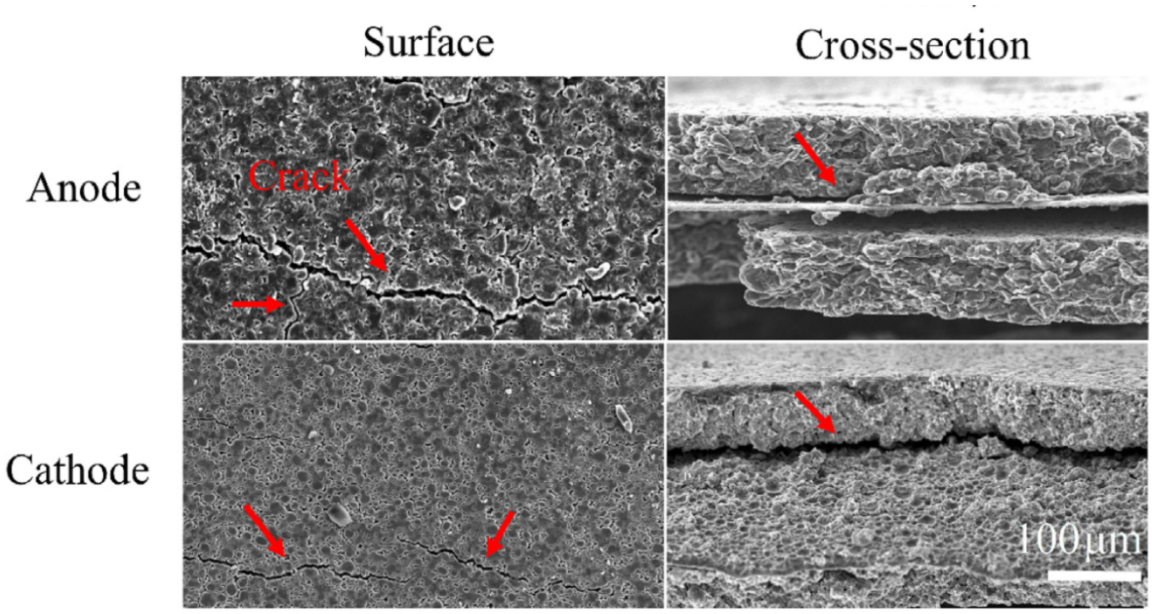


Figure 5: Formation of cracks caused by mechanical stresses in the anode and cathode of LIBs. Reproduced with permission from [65] Copyright 2018, Elsevier.

2.3.2. Degradation mechanisms in the cathode

The degradation of the cathode is caused by TM dissolution, stress generation, particle cracking, and surface layer film formation. All these factors add to the battery aging through degradation.

• *Cathode electrolyte interphase (CEI):* The **CEI** layer forms between the cathode and the electrolyte due to electrolyte oxidation and salt deposition [66]. Unlike SEI, the CEI has low lithium-ion conductivity, and hence the impedance is high. Gas evolution also occurs [67]. LAM and LLI occur at the cathode due to oxidation and subsequent deposition of organic compounds from the anode reduction process. It is difficult to detect the CEI because of the higher cathode voltage [42], [68].

• *Mechanical stress:* During the intercalation and deintercalation of lithium ions, the cathode undergoes a phase transition depending on the battery chemistry, leading to LAM and contact loss **(Figure 5).** Experimental studies indicate that the battery capacity fades at high C-rates due to mechanical stresses generated during lithium intercalation [69].

• *Dissolution of transition metals:* TM oxides are generally used as cathode materials in LIBs [70]. Side reactions occurring at the cathode cause the dissolution of TM ions in the electrolyte during the battery cycling process, especially at high temperatures [71]. The dissolved metal ions are drawn towards the positive electrode and form metallic deposits, leading to dendrite formation. Dendrites affect the battery's functionality by reducing the capacity and safety by developing a short circuit, resulting from piercing the separator. Spinel chemistry in the cathode is particularly sensitive to the TM dissolution effects. The presence of trace amounts of HF in the LiPF electrolyte contributes to the metal dissolution. The dissolution can be minimized by ensuring the electrolyte purity, using dopants, and reducing the cathode-electrolyte interaction using a protective coating.

The above factors collectively result in reduced energy efficiency, capacity loss, and may pose safety hazards, including the risk of thermal runaway. The battery aging mechanism of a single cell is widely investigated in many studies. Battery degradation in EVs affects the performance, driving range, and daily usage. In the EV battery module, the aging of an individual cell affects the overall performance of the EV. In the long run, there is a possibility of accelerating the aging of other cells in the pack. It should be noted that the aging of the battery and the related side reactions are affected by battery design, production, and application. Hence, understanding the battery system degradation is even more complicated than single battery degradation, and the consistent system evolution to the degradation must be accounted for [42]. Degradation of EV batteries can be controlled by some precautions, including reduced exposure to high temperatures in parking locations and avoiding immediate charging after aggressive use, which in turn controls the degradation occurring due to high temperature. Minimizing the operation time at high state of charge (SOC, which is defined as the ratio of its current capacity to the nominal capacity, see Equation (1) in Section 3.1), reducing the maximum charge to SOC to below 90%, and avoiding the EV operation at SOC below15% will also significantly limit the EV battery degradation. Degradation effects due to SOC can also be controlled. Whenever possible, choose slow charging and precondition the battery before use.

Battery SOH and RUL are important parameters to decide the battery application in the SL. Generally, SOH estimation is complicated because it is derived from measurable entities like current and voltage

[72]. In addition, SOH is related to some external parameters like temperature and storage conditions that add to the complexity of accurate estimation. Better SOH estimation is the key to precise RUL prediction. Hence, SOH and RUL are crucial for understanding the degradation and thereby avoiding failure in the FL and SL applications [73].

3. SOH estimation in FLBs and SLBs

3.1. SOH estimation

SOH estimation is significant to determine the battery condition. Battery degradation is a critical parameter influencing the prediction of SOH and depends on the usage and storage conditions in the FL as well as the SL. The battery cycle data is required for interpreting the degradation and estimating the SOH. The data is mostly uncertain, and under certain conditions, confidentiality is maintained by original equipment manufacturers (OEMs). Under such circumstances, good connectivity between the degradation and the available data of the battery is the only way for the SOH estimation for SLB. Estimation of SOH without much experimental and computational time and complexity is advantageous for SLB applications. A better understanding of aging mechanisms leading to battery degradation is fundamental for better analysis. Hence, the information on battery SOC, SOH, and RUL is key information for SLBs.

The SOC of a battery is defined as the ratio of its current capacity (C_t) to the nominal capacity (C_n), in Equation (1):

$$SOC(t) = \frac{C_t}{C_n} \quad (1)$$

where C_n refers to the maximum amount of charge stored in the battery based on the manufacturer's information [74]. It is directly related to the driving range which is an important parameter in EVs [75] SOC evaluation is important for the calculation of the remaining driving range.

SOH estimation methods are yet to be consolidated; they are based on the factors, including capacity, internal resistance, power, and the number of charge/discharge cycles [76], [77], [78]. SOH is defined as the ratio of the maximum available discharge capacity (C_{now}) to that of the nominal capacity (C_n) in Equation (2) [79]:

$$SOH = \frac{C_{now}}{C_n} \times 100\% \quad (2)$$

In terms of power fade (increase in internal energy), SOH is defined in Equation (3),

$$SOH = \frac{R_{EOL} - R_{now}}{R_{EOL} - R_{new}} \times 100\% \quad (3)$$

Here, R_{EOL} refers to the internal resistance at end of life, R_{now} refers to the internal resistance at the current time, and R_{new} refers to the internal resistance of new cell at current time.

Estimation of SOC is required to predict the RUL of a retired battery **[1]**. The RUL can be defined as the number of battery cycles from its rated capacity to the end of its life in Equation (4):

$$RUL = N_{eol} - N \quad (4)$$

where N_{eol} represents the maximum number of battery cycles, and N represents the number of cycles spent by the battery [80].

It is assumed that under normal operating conditions a battery in an EV will last for approximately 8 to 10 years, or the vehicle reaches a driving range of 160,000 km [81]. Either condition, depending on the situation, means the battery capacity drops to 80% of the initial capacity. Electromobility requires high-power operation, and hence, the battery with less than 80% RUL cannot be used in this sector. Such batteries with less power capacity can have a SL in low mobility applications or are placed in grid services.

3.2. SOH estimation methods

The SOH estimation can be classified into direct and indirect approaches, and their detailed methodologies are summarized in **Figure 6**

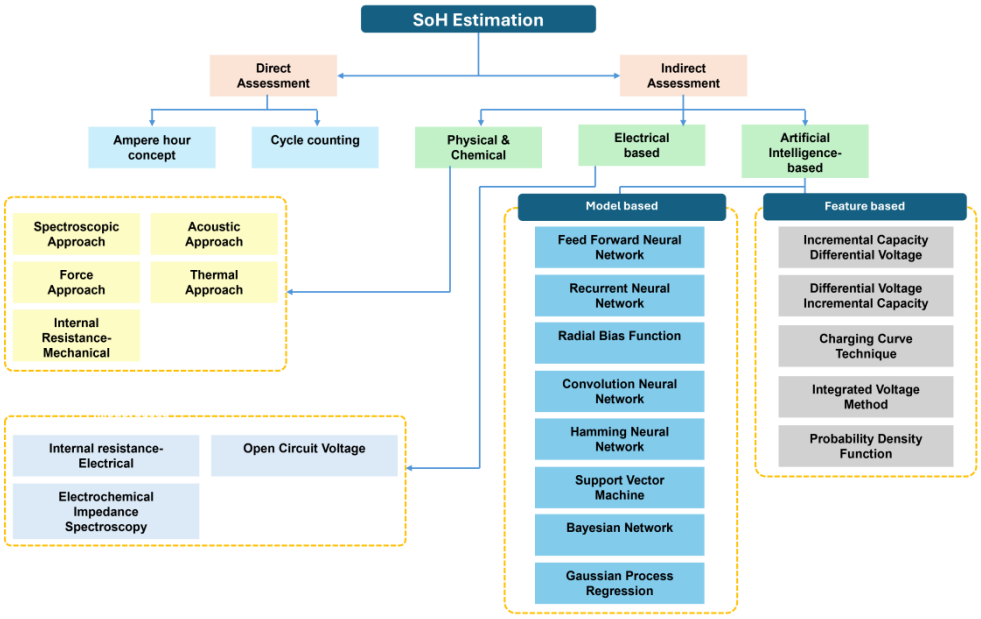


Figure 6: SOH estimation methods. Adapted with permission from [10] Copyright 2024, Elsevier.

3.2.1. Direct methods

Direct methods involve measuring LIB parameters, such as capacity and internal resistance, directly in the laboratory. Ampere-hour and cycle counting are mainly used in the direct assessment approach. The direct methods are accurate and reliable but are not practical when applied to different battery chemistries, large numbers, and varying operating conditions based on the FL application. The tests that determine SOH are intrusive, requiring sophisticated equipment that is highly valued and demands more energy. The process is tedious and removing the LIBs from the current service and performing laboratory tests in real-time are challenging.

3.2.2. Indirect methods

Indirect methodologies encompass physical and chemical-based methods that are destructive in nature, electrical-based techniques that require complex lab equipment to analyze the battery degradation, and lastly, the artificial intelligence (AI)-based data-driven approaches, which require machine learning (ML) or feature-based analysis to articulate the battery degradation mechanism. The battery behaviour is considered a black box, and the equations used do not represent any actual meaning and only mimic it.

• *Data-driven methods:* The complexity of the LIB degradation model is bypassed in the data-driven approach. Data-driven methods work by extracting health indicators of the degradation from LIB experiments and using AI methods like ML or reinforcement to fit the relation between the health indicator and the SOH. Data-driven methods are considered more reliable and precise in estimating SOH compared to other methods. ML models are suitable for different battery types, no prior knowledge of battery operation is necessary, and the additional inputs can be used to train the model based on real-world operating conditions [82]. Techniques like deep neural networks (DNNs) [83], recurrent neural networks (RNNs) [84] , and Gaussian process regression (GPR) [85] can directly map inputs such as current, voltage, and temperature to SOC.

Hybrid models combine the simple techniques used in empirical models for speed and employ the accuracy level of a physics-based approach. The single particle model [86] and the mechanistic model [87] are hybrid models that evaluate cell voltage and capacity based on a given degradation mechanism using an inverse approach. Though direct experimental techniques are simple and robust, the complexity of battery degradation makes them incompatible in real-time applications. Thus, the data-driven approach for evaluation of SOH is adopted because of the complexity of the parameters contributing to the depletion of EV battery and their interconnections. Also, more levels of intricate details should be considered for used EV batteries, as each battery is subjected to different working conditions. Acquiring information on every aspect of the operational condition and degradation of each battery module from EVs is arduous.

The data-driven method used by Jo et al., [88] in the estimation of SOH is shown in **Figure 7**. There are three stages in this method, namely data pre-processing, model training, and performance evaluation.

Data preprocessing: The raw data is cleaned by eliminating missing and abnormal values. The time-based data is converted to SOC-based data. Features are extracted and split into training and test datasets.

Model training: The training dataset is further divided into training and validation datasets. The training dataset is used to train feedforward neural network (FNNs), convolutional neural networks (CNNs), and long short-term memories (LSTMs). The validation dataset is used to adjust the hyper-parameters of the architecture.

Performance evaluation: The test dataset is used to validate the trained model. Performance evaluation is done by calculating the mean absolute error (MAE), mean squared error (MSE), and mean absolute percentage error (MAPE).

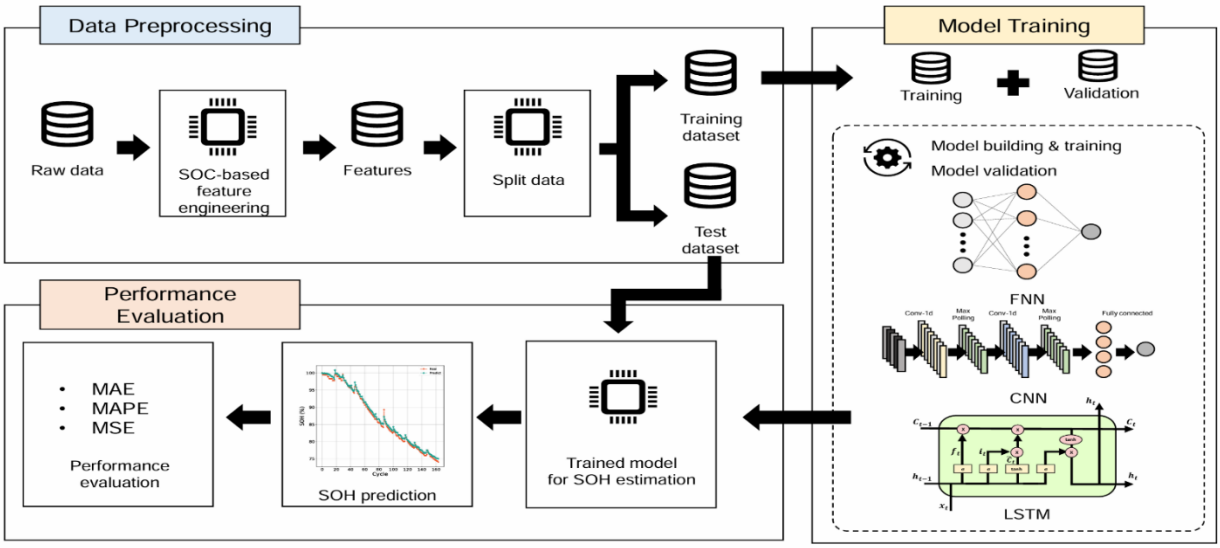


Figure 7: Overview of the process for battery capacity estimation. Reproduced from [88], open access.

3.3. SOH Estimation in FLBs

The FL of EV batteries is characterized by degradation of capacity and power due to internal aging mechanisms, including calendar and cycle aging, as discussed in Sections 2.2-2.3. Side reactions within the battery cell contribute to calendar aging during the idle period of the vehicle. Temperature and SOC influence calendar aging in the battery. Cycle aging is caused by the regular operation [46] and is influenced by factors like temperature, charge/discharge currents, DOD, SOC variation, maximum charge voltage, and total energy throughput. **Figure 8** shows the models used in the study.

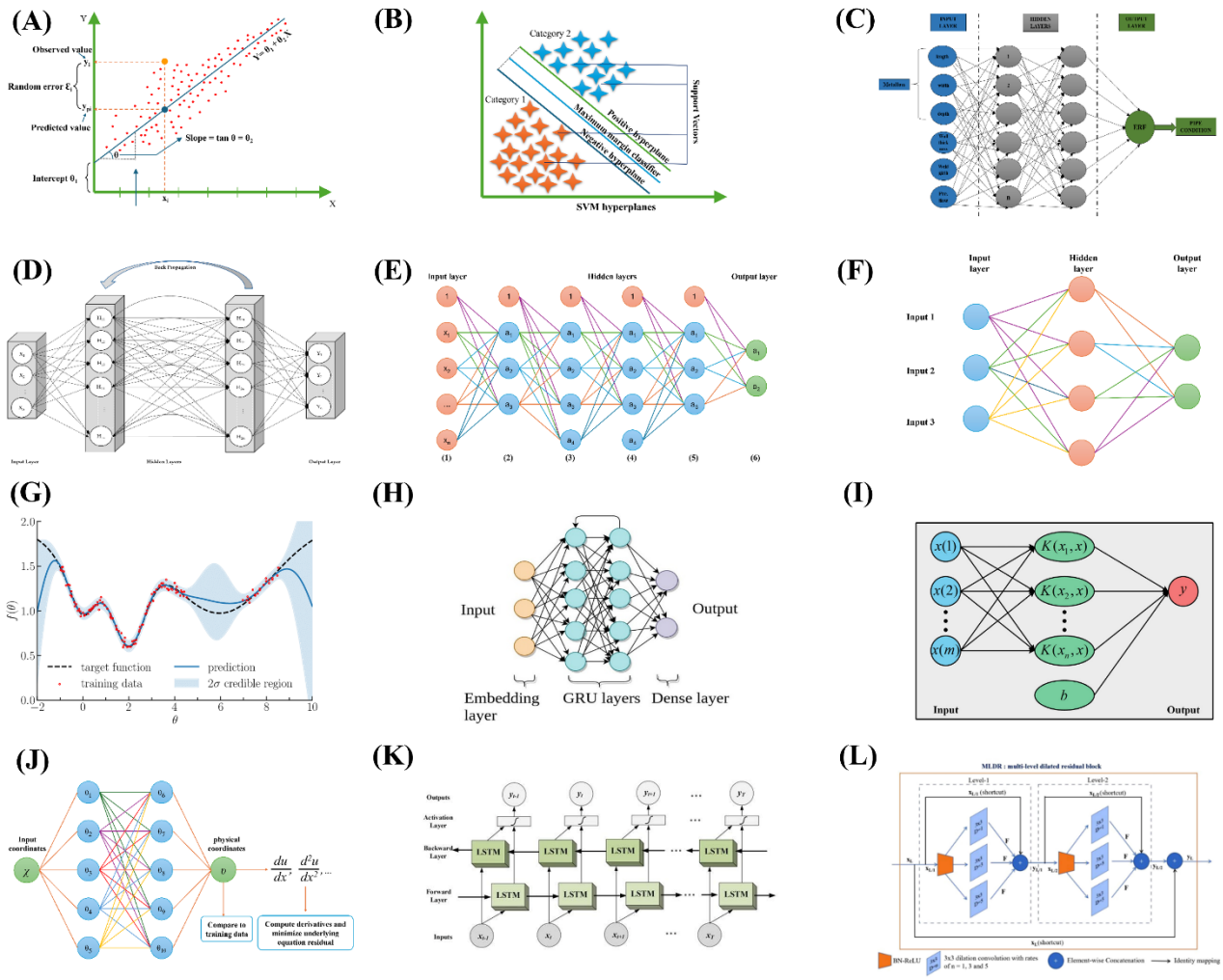


Figure 8: Models used in the SOH estimation. (A) Linear regression, (B) Support Vector M-Kernel method, (C) BPNN -NN [89], (D) NN-LSTM [90], (E) NN -CNN, (F) NN -Feedforward ANN model-MLP, (G) Statistical method-GPR, reproduced with permission [91], (H) NN -GRU. reproduced with permission [92] Copyright 2021, Elsevier, (I) Kernel method- LSSVR, reproduced with permission [93], (J) NN -PINN, (K) NN - BiLSTM neural network, reproduced with permission [94], (L) NN - DRTEM [95].

3.4. Developments of SOH estimation methods in FLBs

This section discusses some recent trends in the SOH estimation of LIBs in their FL. The challenges and improvements in each study are discussed. **Table 1** lists the techniques used for SOH estimation in the discussion. Kashkooli et al., derived the joint estimation of SOC and SOH employing a feedforward artificial neural network (ANN-MLP method). Using aged NMC-NMO pouch cells for SOH estimation, an root mean squared error (RMSE) of 1.67% was achieved, highlighting the importance of ageing data for reliable prediction. For SOC estimation, separate models were applied in the charging and discharging processes to account for the hysteresis effect in LIBs. Notably, the ANNs used in the study can map battery data that include current, voltage, aging time (in months),

storage temperature, and storage condition (charged/discharged) to the current SOC and SOH estimation[96]. The inclusion of diverse battery chemistry, different ageing scenarios, and dynamic operating conditions in the models would lead to better battery management systems (BMSs) for EVs. The enhanced robustness of the tested model, combined with adaptive filtering techniques, could be advantageous for real-time applications when more dynamic conditions and cyclic aging are incorporated. Fang et al., proposed joint online estimation of SOC and SOH in LIBs using an equivalent circuit model (ECM). The online parameters are identified using the forgetting factor recursive least squares (FFRLS) method. The joint online estimation of SOC and SOH is achieved through a dual extended Kalman filter (DEKF). The open circuit voltage (OCV)–SOC curve is reported as the dominant factor influencing estimation accuracy, namely maximum errors of 1.08% SOC and 1.54% SOH[97]. The narrow aging range, computational feasibility for BMS, and SOH estimation based only on resistance are major concerns for real-world applicability. Jiang et al., (2022) systematically compared broadband EIS features **(**full range of frequency data), model parameter features (parameters derived from fitting an EIS model), and fixed-frequency impedance features (data from specific frequency points) for SOH estimation, as shown in **Figure 9A**. The ML method, and GPR were used to estimate SOH based on features. Data from Commercial 18650- LIBs aged under constant and dynamic discharge conditions were used in the study. The fixed frequency EIS data generated better results following the model parameter and broadband EIS features, as shown in **Figure 9B**. The enhanced accuracy and speed make this method suitable for BMS. Comparison of the performance of three EIS features is depicted in **Figure 9C**. This method works well for randomized discharging data and real-world fleet data [98]. Use of EIS data at only one SOC of 50% and temperature of 25 ºC is a limitation for real-world BMS.

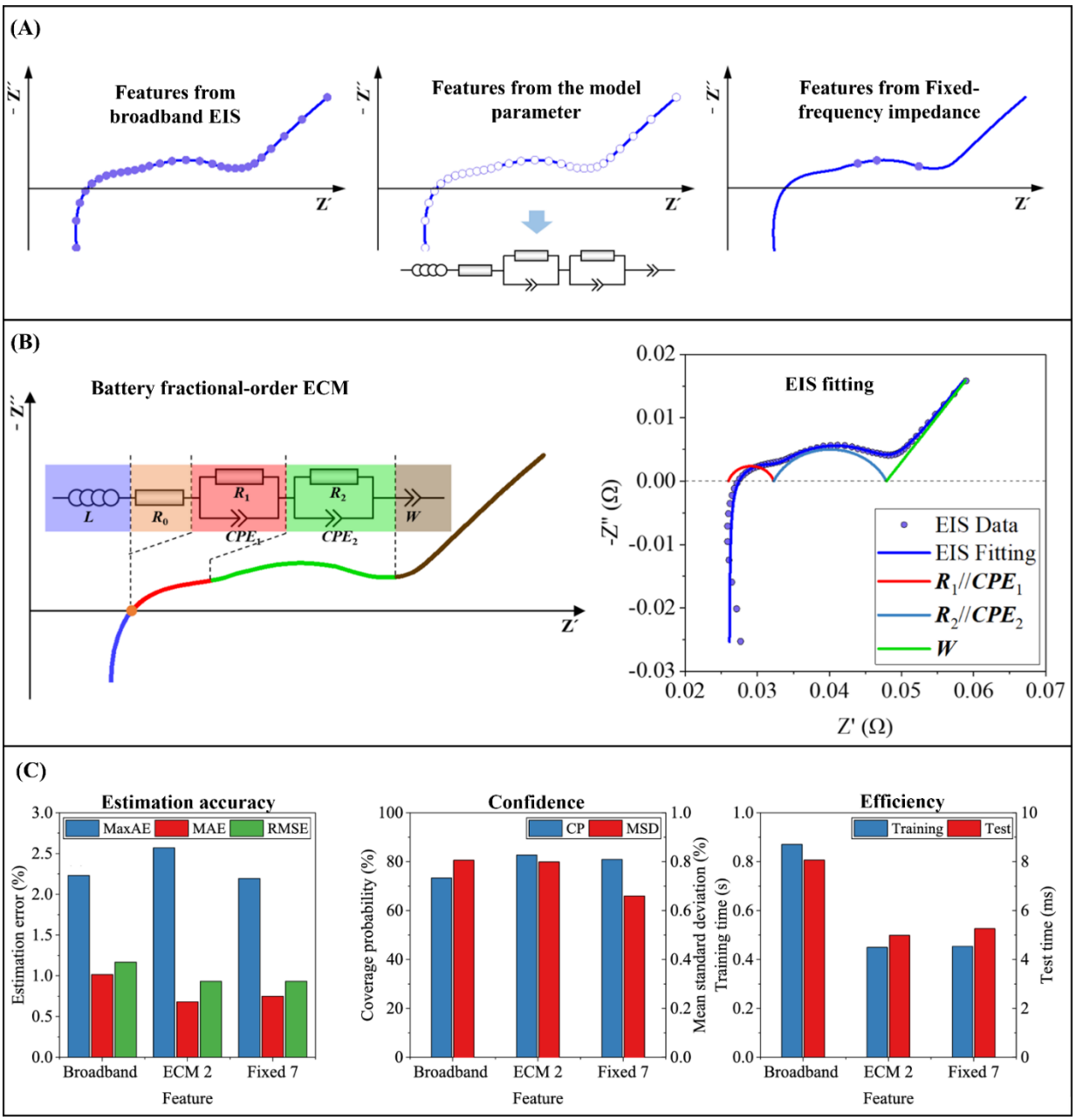


Figure 9: EIS feature based SOH estimation (A) Different feature extraction methods, (B) ECM and EIS fitting, and (C) Performance comparison of three EIS features. Reproduced with permission from [98] Copyright 2022, Elsevier.

Wen et al., proposed a method for the online estimation of SOH for BMS. In this work, SOH at different temperatures is predicted using features from incremental capacity analysis (ICA) applied to a back propagation neural network (BPNN). The IC feature is selected based on the strongest correlation to SOH. IC data are updated using an online correction model to ensure accurate predictions at each aging stage. A low average prediction error of 1.16% is reported [99]. Battery degradation is complicated, while considering only temperature, which is in fact a critical environmental variable for SOH estimation, limits its real-time application.

During charge/discharge cycles, the surface of the cell undergoes strains due to volume changes in the electrode materials [100] and the temperature varies due to internal electrochemical processes [101]. Based on this concept, Li et al., developed a hybrid ML framework which is a pruned CNN and GPR in a feedback loop, by integrating Fiber Bragg grating (FBG) sensors, for SOC and capacity (SOH indicator) estimation. It is found that joint estimation accuracy is improved by the automatic extraction of features from FBG sensors and the feedback loop. The approach significantly improves estimation performance, reducing SOC estimation error by 85.6% and standard deviation by 42.7% [82]. This study is proof of concept for the incorporation of embedded sensors.

Battery HIs were extracted from discharge curves using the U-chord curvature model [102]. The discharge curve is segmented into different phases based on the curvatures, and then health indicators (HIs) are extracted with a strong correlation with SOH. ML models, long short-term memory (LSTM), ANNs such as back propagation feed-forward neural network (BPFFNN), and GPR have been used for SOH estimation of three public datasets. LSTM is good for the initial estimation but underperforms when battery regeneration occurs. Though better prediction was achieved using BPFFNN, GPR performed well overall. The method proves to be effective, reliable, and robust across different battery types. The use of publicly available datasets assures repeatability and reliability in the method.

An accurate estimation method was proposed for SOH under the condition where the battery is not fully charged or discharged by extracting two HIs [103]. The multiscale HI selection strategy used here improves the model framework. The battery capacity estimation indicator is selected based on the initial voltage drop during discharge and the time interval for an equal voltage difference. A novel optimization step is introduced using an elitist genetic algorithm (e-GA) to find the optimal voltage segment. This step maximizes the correlation with the battery capacity for any specific battery, which is a major contribution to the study. A degradation model based on LSTM NNs that were developed using the extracted HIs. An MAPE of 0.41% on average was achieved using this strategy on the public NASA dataset. The computational complexity of pre-processing using e-GA is intense.

A hybrid model was developed to improve the accuracy of SOH estimation of LIBs [104]. The combination of three distinct techniques, empirical mode decomposition (EMD), ICA, and gated recurrent units (GRU), highlights the intelligent combination of signal processing and ML. The EMD-ICA-GRU model utilizes EMD to decompose the capacity regeneration phenomenon and data noise. The HIs are the peaks of ICA curves and the corresponding voltages, which are extracted and used as input to the model. Based on adaptive weighting factors, GRU units are introduced to the combined model. The proposed model was validated with the NASA LIB dataset. An MSE of about 0.3% is reported. Systematic comparison and validation with common NNs like BP, simple RNN, and LSTM add to the fair evaluation of the proposed model.

The battery aging characteristics were studied novel HIs derived from the constant current charging process based on the duration of charging at equal voltage intervals for LFP batteries [93]. The LFP battery possesses a unique challenge of flat OCV characteristics, which complicates monitoring ability. The charging duration of 10 continuous values is obtained from the BMS, and the SOH mapping model is developed based on the least squares support vector regression (LSSVR). Hyperparameters are extracted with particle swarm optimization (PSO), as shown in **Figure 10**. The method is validated using the LFP battery aging dataset. Using a small sample size with high dimensionality, this study was able to achieve better SOH estimation. This method was compared with the NN method of SOH estimation and the results show that the proposed method is better under conditions of small sample data and high-dimensional HIs. To ensure timely and reliable estimation of battery SOH, the authors suggest obtaining the SOH estimation from the discharge process in future studies and combining it with the charging process.

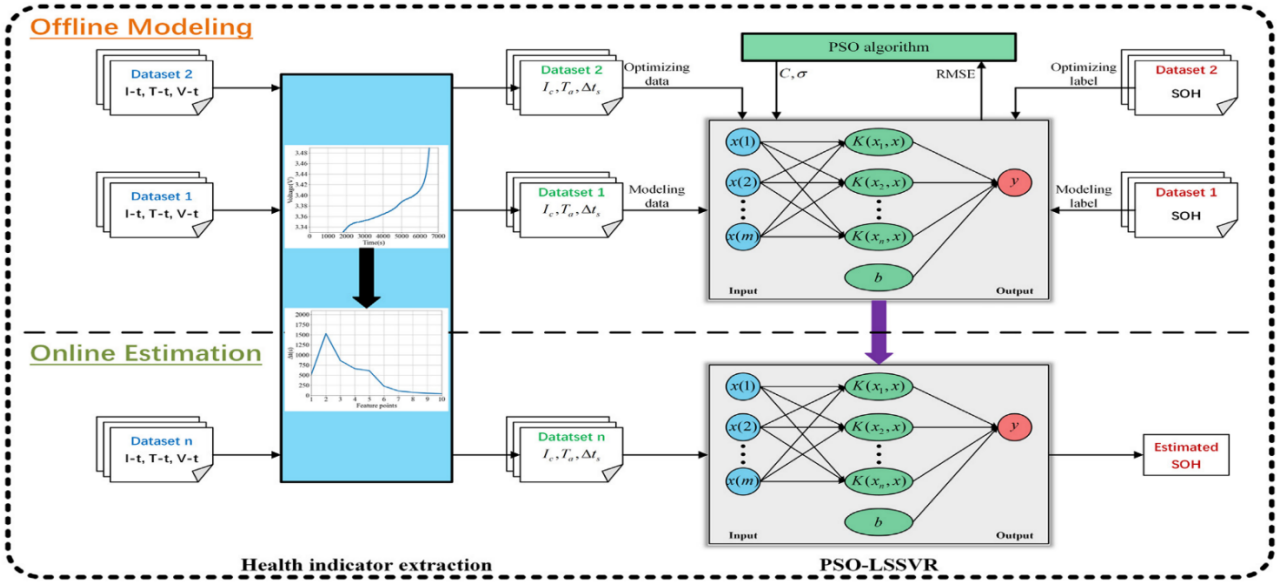


Figure 10: Framework of SOH estimation using HIs and PSO-LSSVR. Reproduced with permission from [93] Copyright 2023, Elsevier.

An advanced hybrid model was introduced based on the physics-informed neural network (PINN) for accurate and stable estimation of SOH of LIBs [105]. Here, empirical degradation models are combined with state-space equations. NNs are used to learn battery degradation patterns. Statistical features from a short segment of charging data are extracted. The features are found to be dependent on the different battery types rather than the charging protocols. The model was validated on large datasets from 387 batteries and 310,705 samples including multiple manufacturers. The proposed method shows a very low MAPE of 0.87% and good performance when small datasets are used and in transfer learning conditions. The high accuracy and versatility of the PINN approach make it a promising method for future battery SOH estimation and degradation modelling.

The SOH was estimated based on a data-driven method employing 61 HIs derived from cycle curves [106]. Discrete wavelet transform (DWT) is used for noise reduction, and a mutual information-based feature selection algorithm refines the indicator set. After considering the correlation and redundancy, 7 HIs are selected and input to bidirectional long short-term memory (BiLSTM) NN models. The method achieves high accuracy, with an MAE below 0.3535% and an MAPE below 0.3747%.

A high-precision voltage prediction method was proposed that works well with sparse data and real-world data for the battery's whole life cycle management [107]. A self-attention network with transfer learning is used to pre-extract features from a sparse dataset, and the voltage can be predicted precisely. Overfitting and distortion issues in training the model with sparse data are tackled by using a new loss function. The training dataset update strategy is incorporated in the model to adapt to address the internal and external changes of the battery. Using experimental and real-world datasets, a MAPE of less than 0.5 % was achieved. This study provides a strong base for a whole life cycle BMS algorithm with accuracy, less data, and low computation.

The correlation between impedance and SOH was studied [108]. This analysis is important to understand the relationship between capacity and impedance of the battery over the whole lifecycle. Feature extraction is done from EIS data by Pearson correlation for the linear relationship of variables and distance correlation for the linear and non-linear relationships of variables. Features from the impedance are extracted depending on the relationship for estimating the SOH. The Pearson correlation shows the highest correlation at SOC of 10%. The results were validated using support vector regression (SVR) and multilayer perception, producing an error ranging 0.84-0.86%. This study emphasizes the importance of feature selection on the accuracy of SOH estimation. The simple model reported in the study produces accurate results with suitable feature extraction.

The relaxation voltage difference is used at different DODs to obtain the SOH and OCV using a combined CNN and BiLSTM [109]. The model, called the multi-gate mixture of experts CNN-BiLSTM (MMoE-CB), was fine-tuned to achieve transfer learning for SOH and OCV. The MAEs for the SOH and OCV were 1.51% and 0.46%, respectively. This method can be improvised by exploring the relation between different RV and SOH. A battery aging dataset was developed under multiple operating conditions [110]. The SOH was estimated using a dilated residual temporal encoding network with multi-head attention (DRTEM). This method was tested and compared with other techniques, showing a good balance between estimation accuracy and computational efficiency.

Table 1: SOH estimation in the FL of LIBs

FLB_SOH_BMS	Experimental/Data	Model/Algorithm	Description	Estimation	Ref.

2019	ECM	DEKF algorithm	OCV-SOC curve is dominant factor influencing estimation accuracy	–	[97]
2022	EIS data	EChM GPR	Comparison of three features from EIS for SOH estimation	–	[98]
2019	GCD cycle data at different C-rates	Feedforward ANN model-MLP.	Separate models for charge and discharge processes.	RMSE of 1.67%	[96]
2022	ICA	BP-NN	IC feature with strong correlation to SOH. Online correction model for accurate prediction	1.16%	[99]
2022	FBG sensors	CNN and GPR	Automatic extraction of features from FBG sensors and feedback loop.	SOC estimation error of 85.6% and SD of 42.7%.	[82]
2022	U-chord curvature model to extract features from the discharge curve	LSTM, BPFFNN, GPR	GPR showing overall robust performance	MAE below 1.08% and RMSE below 1.46%.	[102]
2022	Two HIs extracted from a partially charged battery	LSTM NN	e-GA to maximize correlation with battery capacity	MAPE of 0.41%	[103]
2022	Empirical mode decomposition, ICA curves	GRU	HIs from IC and voltages to input to the model, followed by introducing GRU	MSE of 0.3%	[104]
2023	HIs extracted from constant current charging process.	PSO, LSSVR	With small sample size and high dimensionality	–	[93]
	Statistical features extracted from short segment of charging data	PINN	Feature depending on battery type	MAPE of 0.87%	[105]
2024	HIs derived from cycle curves	BiLSTM NN	DWT for noise reduction, mutual information-based feature selection algorithm to refine the indicator set.	MAE below 0.3535%	[106]
2024	A self-attention network using transfer learning to pre-extract features from sparse dataset	High-precision voltage prediction method	A strategy to refine the training dataset is incorporated in the model to adapt to address the intrinsic and extrinsic battery changes	MAPE less than 0.5 %	[107]
2025	Features extracted from EIS data by Pearson correlation	SVR and multilayer perception	Correlation between impedance and SOH.	0.86% to 0.84%	[108]
2025	Relaxation voltage differences at different DODs	CNN and BiLSTM	Multi-gate Mixture of Experts CNN-BiLSTM (MMoE-CB), was fine-tuned to achieve transfer learning for SOH and OCV.	MAE of 1.51% for SOH and 0.46% for OCV	[109]
2025	Battery dataset with multiple operating conditions	DRTEM	Optimal balance between estimation precision and computational demand	–	[110]

3.5. SOH Estimation in SLBs

It is projected that a sizable number of EV batteries will be available for the SL applications in the near future. EV batteries in SL applications are a favorable choice in terms of economic and

environmental considerations. For retired LIBs to be considered for SL applications, they should fulfill the requirements of safety, reliability, and efficiency as per the SL application. To ensure this, several tests and procedures are necessary to understand their internal conditions and forecast future performance. From the literature, the most sought-after parameter in the test procedures is primarily the SOH of the battery.

The determination of SOH in an SL is more complicated than in FL because of the variability in the driving range and conditions. SOH is considered the most important parameter to decide the suitability of LIBs for SL applications and to assess the degradation process. The residual capacity and energy storage capability, the internal resistance by DC measurement, and the internal impedance by AC measurement reflect the battery's power-delivering capability and are used in the estimation of SOH. Individual parameters or their combined evaluation determine the battery readiness for the SL application [111].

Jiang et al., identified the aging mechanism of LFP SLBs using ICA for estimating battery SOH. The battery capacity is estimated using the regression models, including ordinary least squares, ridge regression, and linear regression (LR) with a correlation-based feature selection method and features like peak height and area extracted from three ICA curves. Pearson's product-moment correlation coefficient is used for the selection of features. The study postulates the contribution of LLI and LAM on the negative electrode in the simultaneous decrease of peaks in IC curves. The estimation error for ordinary least squares and ridge regression was within 2%, while the estimation error was below 3% for LR with a correlation-based feature selection method. The study provides a valuable preliminary investigation on SL EV batteries [112]. Luo et al., proposed a mathematical model with a second-order polynomial function and a high degree of fit (with R^2>0.97) for the rapid evaluation of SOH. Constant phase parameters from the EIS ECM are utilized for the rapid estimation of SOH with excellent accuracy. The prediction error is less than 2%. This method provides a quick assessment methodology for a large volume of decommissioned batteries, as shown in **Figure 11(A&B)** [113].

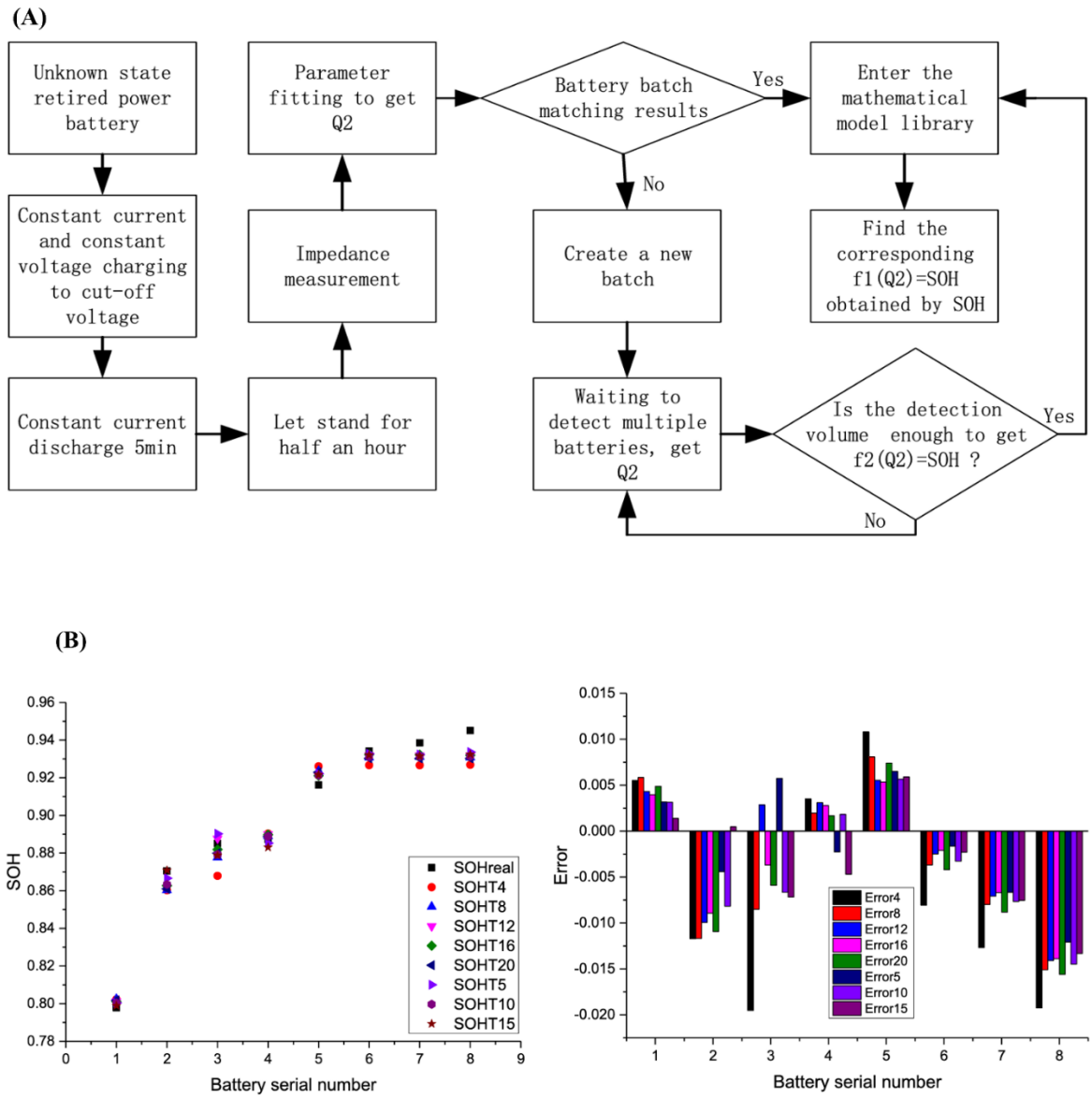


Figure 11: SOH prediction: (A) Block diagram and (B) SOH estimation and error analysis. Reproduced with permission from [113], Copyright 2021, Elsevier.

A method was developed using low/medium-frequency EIS data and GRU NNs, providing a robust, efficient, and scalable solution for estimating the health of retired LIBs [114]. It holds strong potential for practical use in battery recycling and SL applications. The input features of the NN model are the parameters identified by the simplified equivalent circuit model (SECM). The output is the SOH variation curve. Greyscale correlation analysis was used to extract features with high correlation to battery aging. The influence of temperature and SOC on battery health is also considered in the study. RNN, LSTM neutral network, and GRU neutral network are compared in the study. The SECM parameters combined with the GRU NN show precise SOH estimation with smaller estimation error of less than 2% compared to the other two, RNN and LSTM.

The EIS data and a robust ML framework were employed for SOH calculation. GPR is established and verified, and the sensitivity is measured. It is a Bayesian method for non-parametric regression.

GPR works well for small datasets and provides details of the uncertainty values of the predictions. The stability of the method was checked with other cells. The method has been reported with an error of only 1.1% for grading LIBS, as shown in **Figure 12** [115].

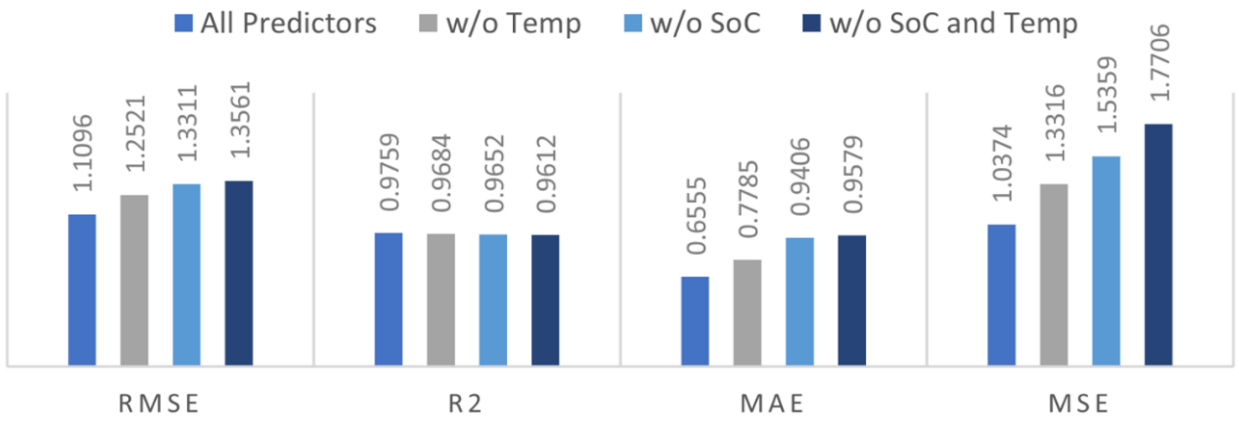


Figure 12: SOC and temperature dependance on SOH prediction. Reproduced with permission from [115] Copyright 2022, Elsevier.

A cross-validation strategy was used to estimate the true SOH by matching the internal resistance and remaining capacity [116]. The basis of resistive remaining capacity (RRC) is introduced in the study to estimate the SOH when SOC is unknown, like in the retired batteries. The traditional method, which uses either remaining capacity or internal resistance, cannot accurately describe SOH. The proposed method combines the extracted feature parameters of retired batteries and reduces the discharge time. The parameters selected for the SOH estimation are voltage and impedance. The fully charged battery is discharged by 10% of battery voltage. After resting the battery for 10 min, both the electrochemical polarization resistance and concentration polarization resistance are measured using a dynamic pulse. After standing for 3 min, the EIS of retired batteries are extracted. **Figure 13A** presents that multiple regression algorithms can take full advantage of the characteristic parameters to accurately estimate the SOH of retired batteries. Among various approaches, the random forest regression (RFR) algorithm that combines multiple decision trees and GPR has the highest accuracy in estimating the SOHC of 11 retired cells. The error in the SOH estimation of 11 batteries is less than 6%. The RRC of the cells is obtained only by internal resistance. **Figure 13B** shows that when the internal resistance of the retired battery deviates from the normal range, the estimated RRC is lower than the actual remaining capacity. This finding suggests that the deviation of RRC from the actual remaining capacity can be used to compare the aging state of the internal resistance of cells with the same SOH.

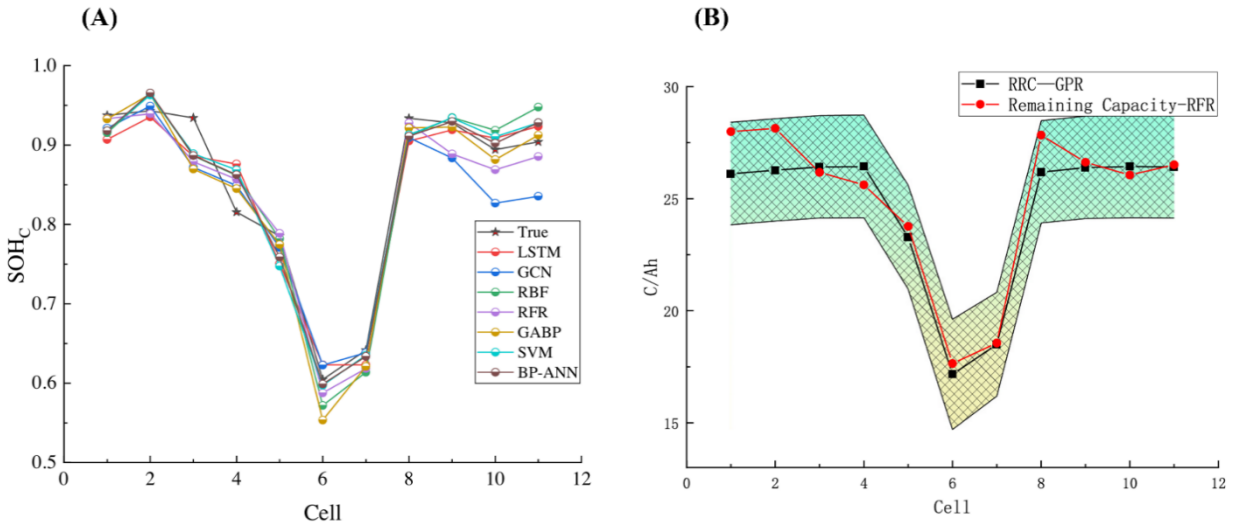


Figure 13: (A) Comparison of multiple regression algorithms for estimating SOH using His; and (B) RRC for GPR estimation with confidence interval of 90 %. Reproduced with permission from [116] Copyright 2023, Elsevier.

Braco et al., proposed a strategy for reducing procedures and cost barriers for SLB application by employing three estimation models based on AC resistance, DC resistance, and the combination of AC and DC resistances in the respective models. This study was conducted on cells, modules, and battery packs from Nissan Leaf vehicles, and the SOH estimation results are given in **Table 2**. The testing time was 2 min (previously 1 day), and the energy consumption was 1 Wh (previously 1.4 kWh) [117].

Table 2: Results of SOH estimation models used in the study [117].

	MAPE (%)		
	Cell (506)	Module (203)	Battery pack (3)
Capacity Prediction	<2.5	<2.5	<4.2
Resistance estimation	<2.4	<2.4	<1.8

An online SOH estimation method was proposed by exploiting the strengths of immediate estimation and long-term forecasting using Kalman filters [118]. In SLB, the frequent estimation of SOH is important as it is more susceptible to external stress than new batteries. Short-term SOH estimation may be inadequate as they are prone to errors. Combining with long-term SOH prediction derived from historic data, the SOH evolution trends can be obtained. GPR is adopted for both short and long-term SOH predictions. This study demonstrates a more economically viable, reliable, and sustainable solution for SLB operation.

Cui et al., introduced innovative offline and online SOH estimation methodology for integration into a SL battery management system for repurposed batteries in grid energy storage applications [119]. For offline SOH estimation, the complete feature extraction process and selection procedure was performed to identify a subset of features that are correlated to SOH. Four ML models based on

online-accessible features and initial capacity are compared, demonstrating the selected model to achieve a MAPE below 2.3% on test data. The online method was based on an adaptive online algorithm which is a clustering-based SOH estimation model. It works along with the offline model based on the elastic net regression (ENR) method to ensure that the estimation error remains within acceptable limits for new, real-time data. This approach aims to promote the safety and durability of SL batteries.

The SOH of LIBs was estimated using experimental EIS data with dimensionality reduction and NNs [111]. Principal component analysis (PCA) was used to suppress the dimensionality without losing useful information. The measurement time of EIS was reduced to 19.0% without losing significant data. Two data sets of 395 LFP cells with an RMSE of 1.18% and 36 NMC cells (0.95%) were used to validate the method offline. This method gives better speed and reduces costs, while preserving battery integrity.

The SOH of Nissan Leaf EV EOL batteries was estimated through non-destructive techniques, providing a practical and comprehensive evaluation of ten non-destructive methods [120]. The His derived in this study apply to the Nissan Leaf 2011 model. The study aims to identify His from advanced techniques rather than the traditional capacity-based approaches, giving rise to deeper insights into battery degradation. The His listed here can be adapted for use with other battery chemistries by the battery repurposer because the non-destructive methods will be useful to characterize SLB when the historic data of the EV is unavailable.

A method was developed to predict the SOH of a SL battery from features extracted from the data recorded in the FL [121]. An SVR model is trained with the FL features and made to estimate the discharge capacity in the SL. SVR is suitable for SOH estimation that deals with non-linear relationships with a small number of trained parameters. The method reduces the complexity and the data acquisition cost by considering the energy delivery of the SLB as the indicator for steady power operation. The RMSE obtained on testing the proposed method ranges from 0.0012 to 0.0158. However, this study did not include the effect of temperature fluctuations on the indicator performance, and more data points from EIS or internal resistance should be incorporated for a more comprehensive approach.

A comprehensive data-driven battery prognostics and health management (PHM) framework was presented [122]. Regardless of the battery operating conditions, PHM provides a precise and robust prediction of SOH and RUL. This study uses a large dataset obtained at different operating conditions for the analysis. His are extracted from partial charging voltage curves that are directly available from the BMS. Non-linear autoregressive with exogenous input (NARX) RNN estimates the SOH using the extracted His. A similarity-based model for RUL forecast is developed with the obtained SOH (predicted value) as the prognostic feature. The model is validated with untrained data. SOH estimations with an RMSE of 0.61 and an RUL MSE of 110 cycles are reported for the method. A

significant limitation is the RUL model's poor accuracy in the early stages of battery life and its reliance on extrapolation to complete the dataset.

A highly practical data-driven method was developed for the rapid assessment of the SOH of retired batteries [123]. FNNs and extra tree regressor (ETR) were tested on the individual cells and modules for the prediction of SOH. The models were trained and tested using experimental measurements from retired batteries. The time and cost for assessing the SOH of the EOL batteries are reduced significantly by adopting these models. MAE less than 1.75% was achieved. The test time is reduced from hours to seconds, and the assessment cost is reduced by 60-82%. The SOH estimation models for SL LIBs discussed above are summarized in **Table 3**.

Table 3: SOH estimation methods in the SL of LIBs.

SLB_ SOH	Experimental/Data	Model/Algorithm	Description	Estimation	Ref
2018	ICA	Ordinary least squares, ridge regression, and LR with a correlation-based feature selection method	Contribution of LLI and LAM to the reduction of peaks in IC curves.	Estimation error within 2% for ordinary least squares and ridge regression; estimation error of LR below 3% with a correlation-based feature selection method	[112]
2021	Constant phase parameters from EIS ECM	Mathematical model with a second-order polynomial function	For a large volume of decommissioned batteries	Prediction error below 2%	[113]
2022	His extracted from partial charging voltage curves	NARX recurrent neural network	Battery PHM framework	RMSE of 0.61, RUL maximum absolute error of 110 cycles	[122]
2023	Low/medium-frequency EIS data	RNN, LSTM, GRU NNs	Influence of temperature and SOC on battery health	GRU NNs showing precise SOH estimation with smaller estimation errors (less than 2%) than RNN and LSTM	[114]
2023	EIS	GPR	GPR useful for small datasets and details of the uncertainty values of the predictions	Only 1.1% error for grading LIBS	[115]
2023	RRC	RFR algorithm combining multiple decision trees and GPR	Feature parameters extracted from retired batteries to reduce discharge time	Less than 6% error	[116]
2023	AC resistance, DC resistance, and the combination of AC and DC resistances		Strategy for reducing procedures and cost barriers	MAPE: cell, module, battery pack capacity prediction:<2.5, <2.5, <4.2, Resistance estimation: <2.4, <2.4, <1.8	[117]

2023		GPR	Short-term estimation and long-term prediction using Kalman filters	Economically viable, reliable, and sustainable solution	[118]
2024		Clustering-based SOH estimation model, ENR	Offline SOH estimation and an adaptive online SOH estimation	Promoting safe and durable operation of SL batteries	[119]
2024	EIS data with dimensionality reduction	NNs	Reduced dimensionality without losing useful information by PCA	395 LFP cells with RMSEs of 1.18% for 395 LFP cells and 0.95% for 36 NMC cells	[111]
2024			Identifying HIs from advanced techniques	Useful method when EV historical data is unavailable	[120]
2024	Features extracted from the FL data	SVR	Reduced complexity and data acquisition cost by energy delivery of SLB as the indicator for steady power operation	RMSE ranging from 0.12 to 1.58%	[121]
2025	Experimental measurements from retired batteries	FNNs and ETR	Reduced test time, reduced assessment cost by 60-82%	MAE lower than 1.75 %	[123]

4. Prediction of RUL and application of SLBs

4.1. Prediction of RUL

The RUL prediction of EV batteries is necessary to ensure the battery's safety and performance. It is a complex challenge, with current methods combining physics-based models and data-driven approaches, particularly ML, and hybrid approaches. The challenges in RUL predictions are mainly due to the differences in the battery data, such as different users and their usage patterns, and changing degradation patterns depending on the FL conditions. The above models analyze factors like charge/discharge cycles, temperature, and usage patterns to provide more accurate RUL predictions than traditional methods. Research is continuously advancing, with new architectures like spatio-temporal attention networks, hybrid models, e.g., combining RNNs with optimization algorithms, and transformer-based models, showing promising results in improving prediction accuracy,

4.1.1. Major methodologies developed to predict RUL of EV batteries

Data-driven approaches: These models use ML to analyze historical data to identify degradation patterns without requiring a deep understanding of the battery's internal physics. For example, researchers are exploring models like LSTM, temporal convolutional networks (TCNs), and extreme gradient boosting (XGBoost) combined with optimization algorithms to improve accuracy. The current review is mainly concerned with this approach.

Physics-based models: These models are based on the underlying physical and chemical processes of the battery. Examples of physics-based models include the double exponential model (DEM) [124], Markov models [125], and two-phase degradation models [126]. The mechanistic calendar aging model by adding the SEI growth reactions to the pseudo-two-dimensional electrochemical model. The electrode degradations predicted by the model were consistent with the dual-tank model and post-mortem analysis [127]. The electrochemical, mechanical, and fracture mechanics to predict battery degradation is discussed in the review by Pistorio et al., [128].

Hybrid approaches: These methods combine the strengths of both physics-based models with data-driven techniques to improve predictions by dynamically adjusting parameters based on observed data, as seen in studies using Coati optimization algorithm to optimize physics-based models. By doing so, hybrid approaches can overcome the limitations of each approach, leading to more robust and accurate predictions. For example, a physics-based calendar aging and cycle aging (CCA) model was combined with an LSTM layer to propose the physics-informed LSTM (PI-LSTM) model for RUL prediction[129]. The physics-informed smooth particle filter (SPF) configuration employs the single particle model (SPM) with SPF algorithm for RUL prediction [130].

4.1.2. Key factors influencing battery RUL

There are many factors that influence battery RUL. Among these, the micro- and nanoscopic battery internal degradation mechanisms in both anodes and cathodes are elaborated in Section 2.3. In this

section, these mechanisms are summarized only briefly, and other external factors such as operational conditions and manufacturing and design aspects are further discussed.

- *Summary of internal degradation mechanisms*

These physical and chemical changes within the battery are the underlying causes of RUL decline:

Formation of SEI: The continuous growth of SEI layer on the anode is one of the most prominent factors leading to capacity fade, as it consumes cyclable Li ions.

LAM: Over time, cracking of electrode particles, corrosion of current collectors, and decomposition of binders lead to a loss of the material essential for storing and delivering energy.

Loss of Li inventory: Side reactions consume the available Li ions, reducing the overall capacity of the battery.

Increase in internal resistance: Degradation processes lead to increased internal resistance, which in turn reduces the battery's ability to deliver power efficiently and generates more heat.

Charging and discharging patterns: DC fast charging accelerates the degradation, causing mechanical stress, especially in hot climate regions. The number of full charge/discharge cycles slowly reduces a battery's maximum potential capacity and battery life. Charging to 100 % (high SOC), complete depletion at discharge (low DOD) conditions will accelerate the degradation [131]. The number of charge and discharge cycles is a primary indicator of the health and the RUL of batteries. Advanced battery technology and BMS can minimise the impact noticeably [132].

Environmental factors: Among many environmental conditions, operating and storage temperatures have a major impact. High temperatures can significantly deteriorate battery health and lifespan, and even cause malfunction. At extremely low temperatures, the battery may cease functioning temporarily and cause issues like lithium plating.

Time: With time, the battery degrades naturally, also termed as calendar aging. This degradation will be accelerated if it remains at high SOC for longer times.

- *Manufacturing and design factors*

Battery chemistry: The specific chemical composition, such as NMC, LFP, LCO and NCA, affects intrinsic performance, longevity, and optimal operating conditions.

Cell balancing: In a multi-cell pack, the weakest cell determines the overall capacity of the battery, meaning that effective cell balancing is crucial for maximum RUL.

BMS: A sophisticated BMS can regulate charge/discharge cycles to stay within optimal parameters, significantly increasing the RUL.

4.1.3. Challenges in RUL prediction

Apart from the above key factors that influence the battery RUL as well as their predictions, there are many other challenges. They include:

• *Complexity:* Battery degradation is influenced by a complex interplay of electrochemical processes and operating conditions. In SL application, the complexity of the degradation process is even greater as the battery operates close to the knee point where performance changes drastically.
• *Non-linear degradation:* Degradation is not always linear and can be irregular, making it difficult for traditional models to predict accurately. Particularly when the battery operates in the non-linear region close to the knee point, the predictability is much less. Predictions are prone to errors when possible, interactions of heterogeneous cells and modules are not accounted for accurately [133].
• *Data availability:* Accurate RUL prediction often requires extensive and high-quality data, which can be challenging to collect and process. The history of the battery, abuse events, or drastic driving patterns will influence the degradation trajectory. Such data are difficult to obtain, and they affect the prediction accuracy. The accuracy of RUL estimation is compromised due to the non-availability of degradation data [1].

4.1.4. Examples of RUL predictions

To address the challenges, much effort has been directed towards developing more reliable ways to predict the RUL of EV batteries, which is crucial for BMSs and for providing consumers with more accurate information about their vehicle's health. Highlighted in the below are some useful approaches developed for RUL prediction of LIBs.

An ANN-based intelligent algorithm for RUL prediction using different training datasets [134]. An advanced BPNN framework derived from a MCI profile is used for RUL prediction of diverse datasets. Voltage, current, and temperature, along with the discharge capacity degradation profile, are used as input parameters. The technique achieved an RMSE of 0.0819. The importance of the capacity degradation curve as a HI is established in terms of capacity regeneration when the battery rests between the charge and discharge processes. The core innovation is the development of MCI to train the BPNN. Further progress in this study is aimed at utilising advanced models like RNN, NARX, and deep learning techniques. Including discharge data, impedance data, and verification based on additional data sources can be beneficial.

A CNN-based method was proposed for online prediction of RUL under realistic and variable operating conditions [135]. The emphasis on the knee point ensures the detection of accelerated degradation early. The ability to handle variable charge/discharge operations strengthens the applicability in real-time. Other notable features in the model include the high accuracy in identifying urgent state batteries with less than 100 cycles before reaching the knee point. Comparison with MLP confirms the CNN's ability to perform well with less historical data. Further testing on different battery chemistries and dynamic conditions will enhance the credibility of the proposed method. In fact, the aging knee is a critical factor for the performance and safety of SLBs. Aging knee refers to the point in the aging curve of a battery beyond which the degradation in battery capacity cannot be reversed, and the battery moves to the EOL situation abruptly [135]. Predicting the aging knee and

preventing it on time can be beneficial for the SLB lifespan as well as predicting RUL of SLBs. Reaching the aging knee can be avoided by improving the battery operating conditions:

- Operation voltage of 80% DOD
- When SOH is below 60%, the operation voltage should be 70% DOD
- Battery temperature 30 °C, operational temperature of 15- 25 °C
- Optimum current below 0.2 C, peak current below 0.3C
- Regular operation of battery. Long-term idleness aggravates calendar aging. SoC and storage condition management can help mitigate calendar ageing [136].

Braco et al., studied the ageing of a SL LIB module from several Nissan Leaf cars through accelerated cycle tests. It aims to determine how internal parameters develop with ageing and derive correlations among them, which are underpinned by curve fitting analyses. A key contribution is the suggestion of an EOL criterion that considers the SL, EOL, capacity, and internal resistance, with the promotion of safety and reliability improvements in real-world applications. The tests indicate that the batteries can perform over 2033 equivalent full cycles before the aging knee point is reached, suggesting an operating life of over 5 years for typical SL applications, such as photovoltaic self-consumption systems. The study also indicates that traditional metrics, such as capacity fading and internal resistance, are insufficient for estimating SL battery life; therefore, more precise ageing diagnostics are required [137].

A hybrid method was proposed based on empirical modelling, capacity recovery correction, and ML [138]. The empirical bi-exponential degradation model provides “Prior knowledge guidance” to the ML model to accurately predict far into the future. The “State recognition and adaptive correction” module uses clustering and support vector machine (SVM) to recognize anomalies in capacity recovery. The capacity recovery correction adapts as per the recoverable capacity with temporary capacity increase during battery degradation. The short-term fluctuations in the training data are corrected to prevent misrepresentation in the long-term forecast. Finally, LSMN utilises the inputs to provide an accurate prediction. The hybrid method ensures better accuracy with a MAPE of just 3.79%, and high reliability with RUL prediction accuracy over 90%. However, the high computational cost and complexity of the model need to be addressed for practical implications.

The suitability of three models, including the semi-empirical model (SeM), ML model and ANN, in predicting the capacity fade and ageing in NMC batteries [139]. Data from real-time ageing of NMC cells were used to test the models. Among these, the ANN model using the non-linear autoregressive network (NARXnet) was found to be most precise in predicting the capacity fade. NARXnet was uniquely capable of capturing complex non-linear phenomena like the sudden degradation "knee point" at the EOL of life and capacity regeneration after resting periods. Based on the outcome, this method is recommended for precision and accuracy in predicting the complex and nonlinear ageing mechanism in NMC-based LIBs.

The RUL of LIB was predicted using a robust hybrid model, whose operational details are schematically presented in **Figure 14** [80]. The study includes complete ensemble empirical mode decomposition with adaptive noise (CEEMDAN) and a bidirectional gated recurrent unit (BiGRU), an improved Grey Wolf Optimizer (IGWO) capable of tuning NN parameters, and validation was done using CALCE and NASA datasets. This method focuses on improved accuracy in prediction, faster optimization, and enriched model generalization across varied frequency components.

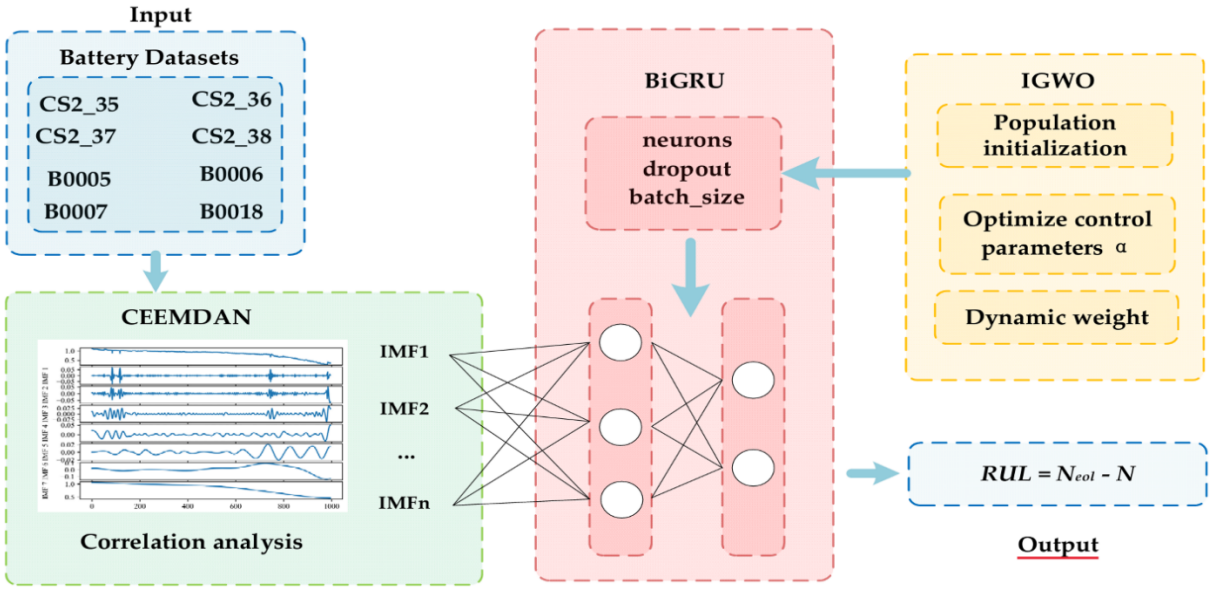


Figure 14: LIB RUL estimation. Reproduced from [80].

The operational details are described in the following:

1. Data Preprocessing: CEEMDAN is used to the collected battery capacity data. The data is decomposed into intrinsic mode functions (IMFs). This step reduces the impact of capacity regeneration.
2. Model Setup: The BiGRU model architecture with input/output/hidden layers is defined. The IGWO parameters, like population size and iterations, are set.
3. Optimisation: The IGWO is used to optimise BiGRU parameters by minimising the RMSE between the actual output and predicted data. IGWO maintains the reliability of the BiGRU network.
4. Prediction: The optimised parameters are applied to the BiGRU model. The IMF components of the battery capacity are predicted.

In another study for accurate prediction of RUL, Xue et al., proposed the combination of an Adaptive Unscented Kalman Filter (AUKF) with genetic algorithm-optimized support vector regression (GA-SVR). The multi-step prediction accuracy is improved by GA-SVR, and the noise in battery data is adjusted by AUKF. The combination of data-driven and model-based techniques is a robust design. NASA battery datasets have been tested by this methodology with better accuracy and reliability for

battery life estimation [140]. In summary, the battery's specifications, including energy and power densities, voltage range, temperature range, drive range, and thermal stability, are crucial factors that determine the overall EV performance. The re-use of EV batteries is regarded as an opportunity to take advantage of the residual energy capacity and to extend the active life of the batteries. Secondary applications proposed for these batteries typically have less energy and power density requirements than their original use in EVs. Nevertheless, there are uncertainties concerning the technical and economic feasibility of these systems. The understanding of the aging processes and the longevity of recycled batteries is important for their SL application. The battery degradation mechanism is very important to estimate the SLB efficiency, reliability, and safety. Early knee point prediction is important in LIBs to ensure safety and optimal usage. The main limitation of models used currently to predict the RUL of LIBs is the requirement for a large dataset. The accuracy is compromised in the case of small datasets. It is helpful when advance models are trained with diverse data to improve the prediction efficiency even when limited training data is used.

4.2. Application of SLBs

The availability of a great number of SL LIBs from used EVs ensures low-cost energy storage solutions for grid storage systems where peak shaving and other ancillary services are possible. The practical challenges of incorporating the SLBs in power systems include variation in the individual cell chemistry and the pack design, lack of standardized methodology for in-field assessment of SOH in FL, and estimation of lifetime expectations in various power systems in SL applications. For the prediction of capacity and power fade in LIBs, model-based approaches are widely adopted as discussed in the previous sections. This section discusses the current trends in SLB applications.

Using new LIBs in grid-connected systems guarantees improved performance, but it is not economically feasible. However, the inclusion of SLBs into grid-connected systems is likely to reduce costs. For the extensive adoption of SLBs in SL applications, it is crucial to precisely handle the complex control and energy management of the large-scale system architecture of SLBs, such as developing an on-site estimation of SOH, and considering the heterogeneous degradation of the SLB pack. As such, more emphasis on reliability and diagnosability is necessary [141]. Here, discussed are a few successful applications of SLBs along with their advantages and feasibility. Using EV LIBs in SL applications is beneficial economically and to the environment. Some pilot-scale projects on SLBs and successful business models are listed in **Table 4**, and the details of a few projects are described in the following.

Table 4: Successful application of SLBs [81]

EV	SLB Projects	Capacity (kWh)	Application	Ref

Volvo XC40	Battery loop energy storage system	-	Solar-powered charging stations (EVs and e-bikes)	[141]
Mercedes Benz	TGN Energy battery energy storage	216	Higher on-site energy consumption	[142]
Audi e-tron	Energy storage system in Herdecke	4500	Frequency stabilization services	[143]
Volvo cars-plug in hybrids	Landafors hydropower plant	250	Regulation of high-speed frequency reserve	[144]
78 Nissan leaf EVs	Enel group's SL project	1700	Provide grid stability	[145]
Jungheinrich (forklift batteries)	Saxony's battery storage facility	-	Provide grid stability and peak shaving	[146]

Volvo cars-plug in hybrids: Fortum has initiated a project to harness SLBs of retired EVs in its hydropower business [147]. This was spurred through a collaborative effort between Fortum's Innovation and Hydropower & Flexibility teams. The project involved collaboration with Volvo Cars, which provided used batteries, and Comsys, whose expertise in the development of battery systems was employed. These SLBs are already being deployed at the Landafors power plant in Sweden, where they enhance the speed and versatility of grid regulation, a critical feature of the management of the kind of fluctuations inherent in renewable sources. Although these batteries can no longer be used in EVs, they retain as long as ten years of RUL in stationary applications. Their application contributes to lowering mechanical stress on turbines via the absorption of regulating load and consequently extends equipment life in the hydropower plant. Moreover, once their SL use has taken place, the batteries remain recyclable, thereby making the project fulfill the requirements of the circular economy.

Zero-emission house (ZEH): SLB offers an affordable option for realizing a ZEH [148], as it costs less than new batteries. It is in line with the sustainability goals of ZEH, reducing waste and emissions. Battery storage will not be opted for in countries with lower electricity prices. A PV-battery system would be expensive. A study on the economic potential of grid-connected residential rooftop PV systems (Türkiye) clarifies that in places where the cost of electricity is less compared to PV-battery utilization, the availability of SLBs makes it economically viable. The cost burden for acquiring a new battery is lowered. SLBs provide frequency regulation and voltage support to grids, while enabling rural electrification [149]. SLBs are useful in supporting grid stability and at times of outages as backup power [150].

Nissan leaf EVs: The microgrid facility installed at the University of California, Davis utilizes SLBs from Nissan Leaf EVs with a capacity of 262 kWh, augmented by solar panels rated at 164.5 kW, to pilot their application for commercial energy storage purposes. The SOH of batteries was determined to be 71% on average, contrary to the general assumption that EV batteries tend to be retired when they are at an SOH of 80%. Throughout the year of operational observation, the microgrid demonstrated a 25% decrease in peak energy consumption and a 23% reduction in peak demand power without noticeable loss of battery performance. This study reveals that SLBs have the potential to perform safely and efficiently in reducing grid demand and increasing the usage of solar power [5]. Reusing used EV batteries as SLBs can store energy and facilitate more renewable energy sources being integrated into the electrical grid, boosting flexibility, stability, and efficiency in the grid. They can also be used in battery storage facilities, electric forklifts, or even ferries, making them an important part of a circular economy, since it extends battery life and tries to diminish waste. From a monetary perspective, SLBs can offer household energy cost savings ranging between 24and 77%, even more if used in conjunction with solar panel systems. Such savings depend on feed-in tariffs (FiTs), which influence the economics of feeding back the energy to the grid storage system versus utilizing it for household use. Also, SLBs assist in shifting peak electricity demand off the grid, thus assisting in lowering GHG emissions due to natural gas power generation. In total, SLBs are environmentally friendly as well as cost-effective while promoting resource efficiency and enhancing grid stability [151]. It should be noted that each SLB is different, and the absence of standardization is a great challenge to the evaluation process. Battery degradation is a crucial parameter for SLBs, and research should focus on ongoing degradation and uncertainty modeling. Gaining in-depth financial and technical risk assessment is pivotal for SLB applications [152]. In short, the adoption of SLBs provides:

- Affordable energy storage systems by supporting smart grid storage and renewable energy sectors.
- Ensure less consumption of new batteries and thereby reduce resource consumption.
- With an increase in SLB adoption, the EV ownership cost can be lowered, increasing the EV usage trend, thereby reducing GHG emissions [153].

5. SLB Implementation, challenges, and outlook

The adoption of technologies like Internet of Things (IoT), data processing, and blockchain improves efficiency, quality, and reliability in customer service in a short time. For selecting an EV battery for second use, it is necessary to analyze several parameters like SOH, RUL, degradation mechanism, battery chemistry, usage history, and economic and environmental feasibility. Numerous data points will be generated and should be evaluated and combined. Developing a business model will require the inclusion of blockchain technology, where there will be a database for references from different manufacturers and end users, or car dealers. Blockchain technology could make the process of

selection and analysis transparent and traceable. Many adverse effects, like market misuse or environmental and safety issues, can be handled systematically. The availability of crucial information will help the effective utilization of energy technology in the long run. The SOH of SLBs in passenger vehicles is unpredictable because of differences in customer usage in terms of charging voltage, routes, and temperatures [154].

5.1. Challenges and perspectives

• *Lack of available data:* It has been iterated that there are significant uncertainties in accurately modelling SLBs and SLB applications because of the highly complex nature and lack of available data for reliable SOH and RUL prediction. In addition, battery performance at different stages of the lifecycle is unpredictable and complicated, the accurate estimation of the essential parameters, like SOH and RUL, is challenging. As such, SLB information and data exchange with energy management systems (EMS) is vital to establish their reliable models as well as for maximum scalability and flexibility.

Optimizing control frames and configuring standard interfaces for SLBs are also part of the holistic approach. The BMSs of SLBs contain sensitive data, such as manufacturer data, user behavior, and usage patterns of batteries, which can be misused if not secured. Such vulnerability has high risks, including:

• *Cyberattacks:* The perpetrators can utilize the charge/discharge cycles of SLBs to induce instability in the power supply or even widespread blackouts.

• *Privacy intrusions:* Personal information such as driving behavior or home addresses can be compromised.

• *Industrial espionage:* The perpetrators can illicitly gain information regarding battery performance and technology, hence violating entities' intellectual property.

Given the above challenging issues, we propose a few potential approaches that can address them in future research.

• *Data Security:* Data security for FL and SL batteries involves protecting sensitive usage history, ensuring transparency for performance trust, managing BMS data, and establishing secure digital identities, primarily addressed through techniques like blockchain, role-based access controls (RBACs), data segregation and standardized DPs to build confidence and comply with regulations. Blockchains create tamper-proof, transparent records of battery usage and degradation, increasing trust in performance data, while the RBACs limit data access only to authorized personnel, crucial for cloud-based management systems and protecting sensitive chemistry or financial data. Data Segregation involves separating customers or facility-specific data for enhanced security.

• *Digital passport (DP):* DP technology refers to a computer system that monitors the entire life cycle of a battery through an embedded identification chip. The system captures critical information

regarding the battery, including date of manufacture, date of beginning of its SL, manufacturer details, material contents, and health data, such as capacity, resistance, and temperature performance. DP technology promotes secure recycling and reuse by enabling all the stakeholders—manufacturers, car makers, recyclers, and consumers alike—entitled to reliable and secure data. From January 2026, the European Commission institutes the use of DPs for EV batteries to promote transparency and sustainability. DP technology enhances the security of battery data and facilitates efficient management of EV batteries throughout their lifecycle [155].

• *Reliable assessment:* The reliable assessment of SOH and the prediction of RUL help all stake holders with the optimization of SLBs in their SL application. The full potential of batteries can be utilized before recycling. By extending the overall lifecycle of EV batteries, the cost and sustainability issues are addressed, which in turn promotes the circular economy. In short, SOH gives current health, and RUL predicts future usability of SLB. Research into accurate estimation and monitoring of SOH and RUL is necessary for the sustainability of SLB applications.

• *Uncertainty modelling:* Through meticulous analysis, a combination of different methods should be adopted to reduce the uncertainty in the RUL prediction of SLBs. Data-driven methods have the advantage of not requiring complex computation and can be adopted for different battery chemistries. However, the reliability of data-driven methods can be elevated by combining them with physics-based models. The tradeoff between reliability, complexity, and affordability is the major challenge facing the SLB industry. With more research and access to data from manufacturers, EV users, a systematic approach should be developed for the seamless operation of the SLB application. Since the SL application is in the early stages of development, more information will be obtained through research and practical applications, which will aid in standardizing the evaluation procedures.

5.2. Conclusion

When considering remanufacture and reuse of a battery in the same application, the whole strategy should be well planned and executed. There are challenges in the recycling process concerning the efficiency of different techniques, pretreatment, transportation, storage, and extraction of valuable metals. Further developments are required for a fully-fledged recycling operation. In view of these issues, repurposing a battery in an SL application is highly beneficial as it bridges the gap between remanufacturing and recycling. The battery is operational for an extended lifetime than the expected lifetime in the EV application. The challenges in the successful implementation of SLBs include the lack of a standardized diagnosis procedure and economic uncertainty because there is no definite explanation or protocol for the EOL of SLBs. There is an urgent need to frame accurate procedures for SOH evaluation and RUL prediction for the successful implementation of SLBs. The process should be reliable, accurate, less time-consuming, and compatible with all EV battery types and conditions. Several research methodologies have emerged, but a solution to all the industrial requirements of SLB has yet to be established. Future investigation should focus on integrating the

different procedures and arrive at a full-fledged model for large-scale applications. Battery analysis is a complex procedure and requires careful integration of relevant technologies. In addition, the incorporation of advanced tools like blockchain and DPs will accelerate progress and increase the credibility of SLB business models.

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Conflict of Interest

The authors declare no conflict of interest.

References

[1] H. Iqbal, S. Sarwar, D. Kirli, J. K. H. Shek, and A. E. Kiprakis, "A survey of second-life batteries based on techno-economic perspective and applications-based analysis," *Carbon Neutrality*, vol. 2, no. 1, p. 8, Apr. 2023, doi: 10.1007/s43979-023-00049-5.

[2] M. Shahjalal *et al.*, "A review on second-life of Li-ion batteries: prospects, challenges, and issues," *Energy*, vol. 241, p. 122881, Feb. 2022, doi: 10.1016/j.energy.2021.122881.

[3] S. Caldani, I. Macchiarelli, B. Unger, and V. Olsen, "Second life: Maximizing lifecycle value of EV batteries," Arthur D. Little. Accessed: Sep. 24, 2025. [Online]. Available: https://www.adlittle.com/en/insights/viewpoints/second-life-maximizing-lifecycle-value-ev-batteries

[4] H. Rallo, G. Benveniste, I. Gestoso, and B. Amante, "Economic analysis of the disassembling activities to the reuse of electric vehicles Li-ion batteries," *Resour. Conserv. Recycl.*, vol. 159, p. 104785, Aug. 2020, doi: 10.1016/j.resconrec.2020.104785.

[5] J. Lacap, J. W. Park, and L. Beslow, "Development and demonstration of microgrid system utilizing second-life electric vehicle batteries," *J. Energy Storage*, vol. 41, p. 102837, Sep. 2021, doi: 10.1016/j.est.2021.102837.

[6] J. W. Lee, M. H. S. M. Haram, G. Ramasamy, S. P. Thiagarajah, E. E. Ngu, and Y. H. Lee, "Technical feasibility and economics of repurposed electric vehicles batteries for power peak shaving," *J. Energy Storage*, vol. 40, p. 102752, Aug. 2021, doi: 10.1016/j.est.2021.102752.

[7] C. White, B. Thompson, and L. G. Swan, "Comparative performance study of electric vehicle batteries repurposed for electricity grid energy arbitrage," *Appl. Energy*, vol. 288, p. 116637, Apr. 2021, doi: 10.1016/j.apenergy.2021.116637.

[8] E. Michelini *et al.*, "Potential and Most Promising Second-Life Applications for Automotive Lithium-Ion Batteries Considering Technical, Economic and Legal Aspects," *Energies (Basel).*, vol. 16, no. 6, p. 2830, Mar. 2023, doi: 10.3390/en16062830.

[9] H. Lee, D. Lim, B. Lee, J. Gu, Y. Choi, and H. Lim, "What is the optimized cost for a used battery?: Economic analysis in case of energy storage system as 2nd life of battery," *J. Clean. Prod.*, vol. 374, p. 133669, Nov. 2022, doi: 10.1016/j.jclepro.2022.133669.

[10] S. Vignesh *et al.*, "State of Health (SoH) estimation methods for second life lithium-ion battery—Review and challenges," *Appl. Energy*, vol. 369, p. 123542, Sep. 2024, doi: 10.1016/j.apenergy.2024.123542.

[11] L. C. Casals, B. Amante García, and C. Canal, "Second life batteries lifespan: Rest of useful life and environmental analysis," *J. Environ. Manage.*, vol. 232, pp. 354–363, Feb. 2019, doi: 10.1016/j.jenvman.2018.11.046.

[12] M. Idris, I. Garniwa, T. E. B. Soesilo, and S. W. Utomo, "Environmental impact on electric vehicle: A cradle-to-cradle approach for various vehicle technologies toward sustainable transportation," *International Journal of Renewable Energy Development*, vol. 15, no. 2, pp. 333–347, Mar. 2026, doi: 10.61435/ijred.2026.61174.

[13] P. Nuss and M. J. Eckelman, "Life Cycle Assessment of Metals: A Scientific Synthesis," *PLoS One*, vol. 9, no. 7, p. e101298, Jul. 2014, doi: 10.1371/journal.pone.0101298.

[14] Amnesty International, "Democratic Republic of Congo: 'This is what we die for': Human rights abuses in the Democratic Republic of the Congo power the global trade in cobalt," Amnesty. Accessed: Nov. 23, 2025. [Online]. Available: https://www.amnesty.org/en/documents/afr62/3183/2016/en/

[15] L. A. W. Ellingsen, G. Majeau-Bettez, B. Singh, A. K. Srivastava, L. O. Valøen, and A. H. Strømman, "Life Cycle Assessment of a Lithium-Ion Battery Vehicle Pack," *J. Ind. Ecol.*, vol. 18, no. 1, pp. 113–124, 2014, doi: 10.1111/jiec.12072.

[16] H. C. Kim, T. J. Wallington, R. Arsenault, C. Bae, S. Ahn, and J. Lee, "Cradle-to-Gate Emissions from a Commercial Electric Vehicle Li-Ion Battery: A Comparative Analysis," *Environ. Sci. Technol.*, vol. 50, no. 14, pp. 7715–7722, Jul. 2016, doi: 10.1021/acs.est.6b00830.

[17] R. Feng, W. Guo, C. Zhang, Y. Nie, and J. Li, "Comparative Study on Environmental Impact of Electric Vehicle Batteries from a Regional and Energy Perspective," *Batteries*, vol. 11, no. 1, p. 23, Jan. 2025, doi: 10.3390/batteries11010023.

[18] H. L. Breetz and D. Salon, "Do electric vehicles need subsidies? Ownership costs for conventional, hybrid, and electric vehicles in 14 U.S. cities," *Energy Policy*, vol. 120, pp. 238–249, Sep. 2018, doi: 10.1016/j.enpol.2018.05.038.

[19] S. P. Holland, E. T. Mansur, N. Z. Muller, and A. J. Yates, "Are There Environmental Benefits from Driving Electric Vehicles? The Importance of Local Factors," *American Economic Review*, vol. 106, no. 12, pp. 3700–3729, Dec. 2016, doi: 10.1257/aer.20150897.

[20] T. R. Hawkins, B. Singh, G. Majeau-Bettez, and A. H. Strømman, "Comparative Environmental Life Cycle Assessment of Conventional and Electric Vehicles," *J. Ind. Ecol.*, vol. 17, no. 1, pp. 53–64, 2013, doi: 10.1111/j.1530-9290.2012.00532.x.

[21] R. Faria, P. Marques, P. Moura, F. Freire, J. Delgado, and A. T. de Almeida, "Impact of the electricity mix and use profile in the life-cycle assessment of electric vehicles," *Renewable and Sustainable Energy Reviews*, vol. 24, pp. 271–287, Aug. 2013, doi: 10.1016/j.rser.2013.03.063.

[22] N. Tilly, T. Yigitcanlar, K. Degirmenci, and A. Paz, "How sustainable is electric vehicle adoption? Insights from a PRISMA review," *Sustain. Cities Soc.*, vol. 117, p. 105950, Dec. 2024, doi: 10.1016/j.scs.2024.105950.

[23] C. Crozier, D. Apostolopoulou, and M. McCulloch, "Mitigating the impact of personal vehicle electrification: A power generation perspective," *Energy Policy*, vol. 118, pp. 474–481, Jul. 2018, doi: 10.1016/j.enpol.2018.03.056.

[24] F. Alanazi, "Electric Vehicles: Benefits, Challenges, and Potential Solutions for Widespread Adaptation," *Applied Sciences*, vol. 13, no. 10, p. 6016, May 2023, doi: 10.3390/app13106016.

[25] J. Neumann *et al.*, "Recycling of Lithium-Ion Batteries—Current State of the Art, Circular Economy, and Next Generation Recycling," *Adv. Energy Mater.*, vol. 12, no. 17, pp. 1–6, May 2022, doi: 10.1002/aenm.202102917.

[26] Q. Dai, J. C. Kelly, L. Gaines, and M. Wang, "Life Cycle Analysis of Lithium-Ion Batteries for Automotive Applications," *Batteries*, vol. 5, no. 2, p. 48, Jun. 2019, doi: 10.3390/batteries5020048.

[27] E. Fan *et al.*, "Sustainable Recycling Technology for Li-Ion Batteries and Beyond: Challenges and Future Prospects," *Chem. Rev.*, vol. 120, no. 14, pp. 7020–7063, Jul. 2020, doi: 10.1021/acs.chemrev.9b00535.

[28] M. L. Machala *et al.*, "Life cycle comparison of industrial-scale lithium-ion battery recycling and mining supply chains," *Nat. Commun.*, vol. 16, no. 1, p. 988, Jan. 2025, doi: 10.1038/s41467-025-56063-x.

[29] G. Harper *et al.*, "Recycling lithium-ion batteries from electric vehicles," *Nature*, vol. 575, no. 7781, pp. 75–86, Nov. 2019, doi: 10.1038/s41586-019-1682-5.

[30] T. Georgi-Maschler, B. Friedrich, R. Weyhe, H. Heegn, and M. Rutz, "Development of a recycling process for Li-ion batteries," *J. Power Sources*, vol. 207, pp. 173–182, Jun. 2012, doi: 10.1016/j.jpowsour.2012.01.152.

[31] A. Chagnes and B. Pospiech, "A brief review on hydrometallurgical technologies for recycling spent lithium-ion batteries," *Journal of Chemical Technology & Biotechnology*, vol. 88, no. 7, pp. 1191–1199, Jul. 2013, doi: 10.1002/jctb.4053.

[32] M. Chen *et al.*, "Recycling End-of-Life Electric Vehicle Lithium-Ion Batteries," *Joule*, vol. 3, no. 11, pp. 2622–2646, Nov. 2019, doi: 10.1016/j.joule.2019.09.014.

[33] S. Al-Asheh, A. Aidan, T. Allawi, F. Hammoud, H. Al Ali, and M. Al Khamiri, "Treatment and recycling of spent lithium-based batteries: a review," *J. Mater. Cycles Waste Manag.*, vol. 26, no. 1, pp. 76–95, Jan. 2024, doi: 10.1007/s10163-023-01842-1.

[34] C. Hendricks, N. Williard, S. Mathew, and M. Pecht, "A failure modes, mechanisms, and effects analysis (FMMEA) of lithium-ion batteries," *J. Power Sources*, vol. 297, pp. 113–120, Nov. 2015, doi: 10.1016/j.jpowsour.2015.07.100.

[35] R. B. Wright *et al.*, "Calendar- and cycle-life studies of advanced technology development program generation 1 lithium-ion batteries," *J. Power Sources*, vol. 110, no. 2, pp. 445–470, Aug. 2002, doi: 10.1016/S0378-7753(02)00210-0.

[36] I. Bloom *et al.*, "An accelerated calendar and cycle life study of Li-ion cells," *J. Power Sources*, vol. 101, no. 2, pp. 238–247, Oct. 2001, doi: 10.1016/S0378-7753(01)00783-2.

[37] S. Barcellona and L. Piegari, “Effect of current on cycle aging of lithium ion batteries,” *J. Energy Storage*, vol. 29, pp. 1–7, Jun. 2020, doi: 10.1016/j.est.2020.101310.

[38] I. Buchberger *et al.*, “Aging Analysis of Graphite/LiNi $_{1/3}$ Mn $_{1/3}$ Co $_{1/3}$ O $_{2}$ Cells Using XRD, PGAA, and AC Impedance,” *J. Electrochem. Soc.*, vol. 162, no. 14, pp. A2737–A2746, Oct. 2015, doi: 10.1149/2.0721514jes.

[39] M. Broussely *et al.*, “Main aging mechanisms in Li ion batteries,” *J. Power Sources*, vol. 146, no. 1–2, pp. 90–96, Aug. 2005, doi: 10.1016/j.jpowsour.2005.03.172.

[40] J. Vetter *et al.*, “Ageing mechanisms in lithium-ion batteries,” *J. Power Sources*, vol. 147, no. 1–2, pp. 269–281, Sep. 2005, doi: 10.1016/j.jpowsour.2005.01.006.

[41] M. Wohlfahrt-Mehrens, C. Vogler, and J. Garche, “Aging mechanisms of lithium cathode materials,” *J. Power Sources*, vol. 127, no. 1–2, pp. 58–64, Mar. 2004, doi: 10.1016/j.jpowsour.2003.09.034.

[42] X. Han *et al.*, “A review on the key issues of the lithium ion battery degradation among the whole life cycle,” *eTransportation*, vol. 1, p. 100005, Aug. 2019, doi: 10.1016/j.etran.2019.100005.

[43] X. Hu, L. Xu, X. Lin, and M. Pecht, “Battery Lifetime Prognostics,” *Joule*, vol. 4, no. 2, pp. 310–346, Feb. 2020, doi: 10.1016/j.joule.2019.11.018.

[44] C. R. Birkl, M. R. Roberts, E. McTurk, P. G. Bruce, and D. A. Howey, “Degradation diagnostics for lithium ion cells,” *J. Power Sources*, vol. 341, pp. 373–386, Feb. 2017, doi: 10.1016/j.jpowsour.2016.12.011.

[45] Y. Miao, Y. Gao, X. Liu, Y. Liang, and L. Liu, “Analysis of State-of-Charge Estimation Methods for Li-Ion Batteries Considering Wide Temperature Range,” *Energies (Basel).*, vol. 18, no. 5, p. 1188, Feb. 2025, doi: 10.3390/en18051188.

[46] A. Maheshwari, M. Heck, and M. Santarelli, “Cycle aging studies of lithium nickel manganese cobalt oxide-based batteries using electrochemical impedance spectroscopy,” *Electrochim. Acta*, vol. 273, pp. 335–348, May 2018, doi: 10.1016/j.electacta.2018.04.045.

[47] T. Rahman and T. Alharbi, “Exploring Lithium-Ion Battery Degradation: A Concise Review of Critical Factors, Impacts, Data-Driven Degradation Estimation Techniques, and Sustainable Directions for Energy Storage Systems,” *Batteries*, vol. 10, no. 7, p. 220, Jun. 2024, doi: 10.3390/batteries10070220.

[48] M. Lucu *et al.*, “Data-driven nonparametric Li-ion battery ageing model aiming at learning from real operation data - Part B: Cycling operation,” *J. Energy Storage*, vol. 30, p. 101410, Aug. 2020, doi: 10.1016/j.est.2020.101410.

[49] P. Huang, G. Zeng, Y. He, S. Liu, E. Li, and Z. Bai, “Damage evolution mechanism and early warning using long short-term memory networks for battery slight overcharge cycles,” *Renew. Energy*, vol. 217, p. 119171, Nov. 2023, doi: 10.1016/j.renene.2023.119171.

[50] M. Liang, B. Luo, and L. Zhi, “Application of graphene and graphene-based materials in clean energy-related devices,” *Int. J. Energy Res.*, vol. 33, no. 13, pp. 1161–1170, Oct. 2009, doi: 10.1002/er.1598.

[51] L. Spitthoff, M. S. Wahl, J. J. Lamb, P. R. Shearing, P. J. S. Vie, and O. S. Burheim, “On the Relations between Lithium-Ion Battery Reaction Entropy, Surface Temperatures and Degradation,” *Batteries*, vol. 9, no. 5, p. 249, Apr. 2023, doi: 10.3390/batteries9050249.

[52] R. Narayan, C. Laberty-Robert, J. Pelta, J. Tarascon, and R. Dominko, "Self-Healing: An Emerging Technology for Next-Generation Smart Batteries," *Adv. Energy Mater.*, vol. 12, no. 17, May 2022, doi: 10.1002/aenm.202102652.

[53] M. Kaliaperumal *et al.*, "Cause and Mitigation of Lithium-Ion Battery Failure—A Review," *Materials*, vol. 14, no. 19, p. 5676, Sep. 2021, doi: 10.3390/ma14195676.

[54] J. Sun, L. Ye, X. Zhao, P. Zhang, and J. Yang, "Electronic Modulation and Structural Engineering of Carbon-Based Anodes for Low-Temperature Lithium-Ion Batteries: A Review," *Molecules*, vol. 28, no. 5, p. 2108, Feb. 2023, doi: 10.3390/molecules28052108.

[55] T. Ohzuku, Y. Iwakoshi, and K. Sawai, "Formation of Lithium-Graphite Intercalation Compounds in Nonaqueous Electrolytes and Their Application as a Negative Electrode for a Lithium Ion (Shuttlecock) Cell," *J. Electrochem. Soc.*, vol. 140, no. 9, p. 2490, 1993, doi: 10.1149/1.2220849.

[56] A. Barré, B. Deguilhem, S. Grolleau, M. Gérard, F. Suard, and D. Riu, "A review on lithium-ion battery ageing mechanisms and estimations for automotive applications," *J. Power Sources*, vol. 241, pp. 680–689, Nov. 2013, doi: 10.1016/j.jpowsour.2013.05.040.

[57] A. J. Smith *et al.*, "Localized lithium plating under mild cycling conditions in high-energy lithium-ion batteries," *J. Power Sources*, vol. 573, p. 233118, Jul. 2023, doi: 10.1016/j.jpowsour.2023.233118.

[58] R. Xiong, Y. Pan, W. Shen, H. Li, and F. Sun, "Lithium-ion battery aging mechanisms and diagnosis method for automotive applications: Recent advances and perspectives," *Renewable and Sustainable Energy Reviews*, vol. 131, p. 110048, Oct. 2020, doi: 10.1016/j.rser.2020.110048.

[59] G. Liu *et al.*, "Controllable long-term lithium replenishment for enhancing energy density and cycle life of lithium-ion batteries," *Energy Environ. Sci.*, vol. 17, no. 3, pp. 1163–1174, 2024, doi: 10.1039/D3EE03740A.

[60] J. Gu *et al.*, "Decoding Internal Stress-Induced Micro-Short Circuit Events in Sulfide-Based All-Solid-State Li-Metal Batteries via Operando Pressure Measurements," *Adv. Energy Mater.*, vol. 13, no. 47, p. 2302643, Dec. 2023, doi: 10.1002/aenm.202302643.

[61] H. Wang and J. F. Whitacre, "Inhomogeneous aging of cathode materials in commercial 18650 lithium ion battery cells," *J. Energy Storage*, vol. 35, p. 102244, Mar. 2021, doi: 10.1016/j.est.2021.102244.

[62] H. Fathiannasab, L. Zhu, and Z. Chen, "Chemo-mechanical modeling of stress evolution in all-solid-state lithium-ion batteries using synchrotron transmission X-ray microscopy tomography," *J. Power Sources*, vol. 483, p. 229028, Jan. 2021, doi: 10.1016/j.jpowsour.2020.229028.

[63] V. Agubra and J. Fergus, "Lithium Ion Battery Anode Aging Mechanisms," *Materials*, vol. 6, no. 4, pp. 1310–1325, Mar. 2013, doi: 10.3390/ma6041310.

[64] J. Xu, "Critical Review on cathode–electrolyte Interphase Toward High-Voltage Cathodes for Li-Ion Batteries," *Nanomicro Lett.*, vol. 14, no. 1, p. 166, Dec. 2022, doi: 10.1007/s40820-022-00917-2.

[65] L. Wang, S. Yin, C. Zhang, Y. Huan, and J. Xu, "Mechanical characterization and modeling for anodes and cathodes in lithium-ion batteries," *J. Power Sources*, vol. 392, pp. 265–273, Jul. 2018, doi: 10.1016/j.jpowsour.2018.05.007.

[66] C. Shinagawa, H. Ushiyama, and K. Yamashita, “Multiscale Simulations for Lithium-Ion Batteries: SEI Film Growth and Capacity Fading,” *J. Electrochem. Soc.*, vol. 164, no. 13, pp. A3018–A3024, Oct. 2017, doi: 10.1149/2.0381713jes.

[67] L. Wang *et al.*, “Research progress of the electrochemical impedance technique applied to the high-capacity lithium-ion battery,” *International Journal of Minerals, Metallurgy and Materials*, vol. 28, no. 4, pp. 538–552, Apr. 2021, doi: 10.1007/s12613-020-2218-6.

[68] M. Kemeny, P. Ondrejka, and M. Mikolasek, “Comprehensive Degradation Analysis of NCA Li-Ion Batteries via Methods of Electrochemical Characterisation for Various Stress-Inducing Scenarios,” *Batteries*, vol. 9, no. 1, p. 33, Jan. 2023, doi: 10.3390/batteries9010033.

[69] C. Zhan, T. Wu, J. Lu, and K. Amine, “Dissolution, migration, and deposition of transition metal ions in Li-ion batteries exemplified by Mn-based cathodes – a critical review,” *Energy Environ. Sci.*, vol. 11, no. 2, pp. 243–257, 2018, doi: 10.1039/C7EE03122J.

[70] X. Liao *et al.*, “Understanding self-discharge mechanism of layered nickel cobalt manganese oxide at high potential,” *J. Power Sources*, vol. 286, pp. 551–556, Jul. 2015, doi: 10.1016/j.jpowsour.2015.04.032.

[71] J. C. Hunter, “Preparation of a new crystal form of manganese dioxide: λ-MnO2,” *J. Solid State Chem.*, vol. 39, no. 2, pp. 142–147, Sep. 1981, doi: 10.1016/0022-4596(81)90323-6.

[72] L. Yao *et al.*, “A review of lithium-ion battery state of health estimation and prediction methods,” *World Electric Vehicle Journal*, vol. 12, no. 3, Sep. 2021, doi: 10.3390/wevj12030113.

[73] H. M. Fahmy, H. M. Hasanien, I. Alsaleh, H. Ji, and A. Alassaf, “State of health estimation of lithium-ion battery using dual adaptive unscented Kalman filter and Coulomb counting approach,” *J. Energy Storage*, vol. 88, p. 111557, May 2024, doi: 10.1016/j.est.2024.111557.

[74] W. Y. Chang, “The State of Charge Estimating Methods for Battery: A Review,” *ISRN Applied Mathematics*, vol. 2013, pp. 1–7, Jul. 2013, doi: 10.1155/2013/953792.

[75] H. Wei, C. He, J. Li, and L. Zhao, “Online estimation of driving range for battery electric vehicles based on SOC-segmented actual driving cycle,” *J. Energy Storage*, vol. 49, p. 104091, May 2022, doi: 10.1016/j.est.2022.104091.

[76] J. Remmlinger, M. Buchholz, M. Meiler, P. Bernreuter, and K. Dietmayer, “State-of-health monitoring of lithium-ion batteries in electric vehicles by on-board internal resistance estimation,” *J. Power Sources*, vol. 196, no. 12, pp. 5357–5363, Jun. 2011, doi: 10.1016/j.jpowsour.2010.08.035.

[77] S. Yang, C. Zhang, J. Jiang, W. Zhang, L. Zhang, and Y. Wang, “Review on state-of-health of lithium-ion batteries: Characterizations, estimations and applications,” *J. Clean. Prod.*, vol. 314, p. 128015, Sep. 2021, doi: 10.1016/j.jclepro.2021.128015.

[78] H. Tian, P. Qin, K. Li, and Z. Zhao, “A review of the state of health for lithium-ion batteries: Research status and suggestions,” *J. Clean. Prod.*, vol. 261, p. 120813, Jul. 2020, doi: 10.1016/j.jclepro.2020.120813.

[79] Y. Fan, F. Xiao, C. Li, G. Yang, and X. Tang, “A novel deep learning framework for state of health estimation of lithium-ion battery,” *J. Energy Storage*, vol. 32, p. 101741, Dec. 2020, doi: 10.1016/j.est.2020.101741.

[80] X. Tang, H. Wan, W. Wang, M. Gu, L. Wang, and L. Gan, "Lithium-Ion Battery Remaining Useful Life Prediction Based on Hybrid Model," *Sustainability*, vol. 15, no. 7, p. 6261, Apr. 2023, doi: 10.3390/su15076261.

[81] K. Nováková, A. Pražanová, D.-I. Stroe, and V. Knap, "Second-Life of Lithium-Ion Batteries from Electric Vehicles: Concept, Aging, Testing, and Applications," *Energies (Basel).*, vol. 16, no. 5, p. 2345, Feb. 2023, doi: 10.3390/en16052345.

[82] Y. Li *et al.*, "A hybrid machine learning framework for joint SOC and SOH estimation of lithium-ion batteries assisted with fiber sensor measurements," *Appl. Energy*, vol. 325, p. 119787, Nov. 2022, doi: 10.1016/j.apenergy.2022.119787.

[83] E. Chemali, P. J. Kollmeyer, M. Preindl, and A. Emadi, "State-of-charge estimation of Li-ion batteries using deep neural networks: A machine learning approach," *J. Power Sources*, vol. 400, pp. 242–255, Oct. 2018, doi: 10.1016/j.jpowsour.2018.06.104.

[84] E. Chemali, P. J. Kollmeyer, M. Preindl, R. Ahmed, and A. Emadi, "Long Short-Term Memory Networks for Accurate State-of-Charge Estimation of Li-ion Batteries," *IEEE Transactions on Industrial Electronics*, vol. 65, no. 8, pp. 6730–6739, Aug. 2018, doi: 10.1109/TIE.2017.2787586.

[85] G. O. Sahinoglu, M. Pajovic, Z. Sahinoglu, Y. Wang, P. V. Orlik, and T. Wada, "Battery State-of-Charge Estimation Based on Regular/Recurrent Gaussian Process Regression," *IEEE Transactions on Industrial Electronics*, vol. 65, no. 5, pp. 4311–4321, May 2018, doi: 10.1109/TIE.2017.2764869.

[86] J. Tian, Y. Wang, and Z. Chen, "An improved single particle model for lithium-ion batteries based on main stress factor compensation," *J. Clean. Prod.*, vol. 278, p. 123456, Jan. 2021, doi: 10.1016/j.jclepro.2020.123456.

[87] M. Dubarry, C. Truchot, and B. Y. Liaw, "Synthesize battery degradation modes via a diagnostic and prognostic model," *J. Power Sources*, vol. 219, pp. 204–216, Dec. 2012, doi: 10.1016/j.jpowsour.2012.07.016.

[88] S. Jo, S. Jung, and T. Roh, "Battery State-of-Health Estimation Using Machine Learning and Preprocessing with Relative State-of-Charge," *Energies (Basel).*, vol. 14, no. 21, p. 7206, Nov. 2021, doi: 10.3390/en14217206.

[89] N. B. Shaik, S. R. Pedapati, S. A. A. Taqvi, A. R. Othman, and F. A. A. Dzubir, "A Feed-Forward Back Propagation Neural Network Approach to Predict the Life Condition of Crude Oil Pipeline," *Processes*, vol. 8, no. 6, p. 661, Jun. 2020, doi: 10.3390/pr8060661.

[90] P. S. Muhuri, P. Chatterjee, X. Yuan, K. Roy, and A. Esterline, "Using a Long Short-Term Memory Recurrent Neural Network (LSTM-RNN) to Classify Network Attacks," *Information*, vol. 11, no. 5, p. 243, May 2020, doi: 10.3390/info11050243.

[91] F. Leclercq, "Bayesian optimization for likelihood-free cosmological inference," *Physical Review D*, vol. 98, no. 6, p. 063511, Sep. 2018, doi: 10.1103/PhysRevD.98.063511.

[92] A. Kumar, H. Rasche, B. Grüning, and R. Backofen, "Tool recommender system in Galaxy using deep learning," *Gigascience*, vol. 10, no. 1, Jan. 2021, doi: 10.1093/gigascience/giaa152.

[93] J. Chen, Y. Hu, Q. Zhu, H. Rashid, and H. Li, "A novel battery health indicator and PSO-LSSVR for LiFePO4 battery SOH estimation during constant current charging," *Energy*, vol. 282, p. 128782, Nov. 2023, doi: 10.1016/j.energy.2023.128782.

[94] F. Zhang, H. Fleyeh, and C. Bales, "A hybrid model based on bidirectional long short-term memory neural network and Catboost for short-term electricity spot price forecasting," *Journal of the Operational Research Society*, vol. 73, no. 2, pp. 301–325, Feb. 2022, doi: 10.1080/01605682.2020.1843976.

[95] N. R. Gudhe *et al.*, "Multi-level dilated residual network for biomedical image segmentation," *Sci. Rep.*, vol. 11, no. 1, p. 14105, Jul. 2021, doi: 10.1038/s41598-021-93169-w.

[96] A. G. Kashkooli, H. Fathiannasab, Z. Mao, and Z. Chen, "Application of Artificial Intelligence to State-of-Charge and State-of-Health Estimation of Calendar-Aged Lithium-Ion Pouch Cells," *J. Electrochem. Soc.*, vol. 166, no. 4, pp. A605–A615, 2019, doi: 10.1149/2.0411904jes.

[97] L. Fang, J. Li, and B. Peng, "Online Estimation and Error Analysis of both SOC and SOH of Lithium-ion Battery based on DEKF Method," *Energy Procedia*, vol. 158, pp. 3008–3013, Feb. 2019, doi: 10.1016/j.egypro.2019.01.974.

[98] B. Jiang, J. Zhu, X. Wang, X. Wei, W. Shang, and H. Dai, "A comparative study of different features extracted from electrochemical impedance spectroscopy in state of health estimation for lithium-ion batteries," *Appl. Energy*, vol. 322, p. 119502, Sep. 2022, doi: 10.1016/j.apenergy.2022.119502.

[99] J. Wen, X. Chen, X. Li, and Y. Li, "SOH prediction of lithium battery based on IC curve feature and BP neural network," *Energy*, vol. 261, p. 125234, Dec. 2022, doi: 10.1016/j.energy.2022.125234.

[100] L. W. Sommer *et al.*, "Monitoring of Intercalation Stages in Lithium-Ion Cells over Charge-Discharge Cycles with Fiber Optic Sensors," *J. Electrochem. Soc.*, vol. 162, no. 14, pp. A2664–A2669, Oct. 2015, doi: 10.1149/2.0361514jes.

[101] C. Zhang, K. Li, and J. Deng, "Real-time estimation of battery internal temperature based on a simplified thermoelectric model," *J. Power Sources*, vol. 302, pp. 146–154, Jan. 2016, doi: 10.1016/j.jpowsour.2015.10.052.

[102] H. H. Goh, Z. Lan, D. Zhang, W. Dai, T. A. Kurniawan, and K. C. Goh, "Estimation of the state of health (SOH) of batteries using discrete curvature feature extraction," *J. Energy Storage*, vol. 50, p. 104646, Jun. 2022, doi: 10.1016/j.est.2022.104646.

[103] Z. Yu, Y. Zhang, L. Qi, and R. Li, "SOH Estimation Method for Lithium-ion Battery Based on Discharge Characteristics," *Int. J. Electrochem. Sci.*, vol. 17, no. 7, p. 220725, Jul. 2022, doi: 10.20964/2022.07.38.

[104] H. Sun, D. Yang, J. Du, P. Li, and K. Wang, "Prediction of Li-ion battery state of health based on data-driven algorithm," *Energy Reports*, vol. 8, pp. 442–449, Nov. 2022, doi: 10.1016/j.egyr.2022.11.134.

[105] F. Wang, Z. Zhai, Z. Zhao, Y. Di, and X. Chen, "Physics-informed neural network for lithium-ion battery degradation stable modeling and prognosis," *Nat. Commun.*, vol. 15, no. 1, p. 4332, May 2024, doi: 10.1038/s41467-024-48779-z.

[106] T. Tao *et al.*, "Data-based health indicator extraction for battery SOH estimation via deep learning," *J. Energy Storage*, vol. 78, p. 109982, Feb. 2024, doi: 10.1016/j.est.2023.109982.

[107] J. Hong *et al.*, "Online accurate voltage prediction with sparse data for the whole life cycle of Lithium-ion batteries in electric vehicles," *Appl. Energy*, vol. 369, p. 123600, Sep. 2024, doi: 10.1016/j.apenergy.2024.123600.

[108] S. Pohlmann *et al.*, "Correlation analysis and feature extraction using impedance spectroscopy over aging of lithium ion batteries," *J. Energy Storage*, vol. 105, p. 114715, Jan. 2025, doi: 10.1016/j.est.2024.114715.

[109] Y. Liu, Y. Liu, X. Bao, and H. Shen, "Estimation of battery state of health and open circuit voltage at various depths of discharge based on deep learning and relaxation voltage," *Energy*, vol. 328, p. 136555, Aug. 2025, doi: 10.1016/j.energy.2025.136555.

[110] S. Yang, H. Xie, S. Liu, Y. Fan, and X. Tan, "A method for estimating the SOH of lithium-ion batteries under complex charging conditions using dilated residual temporal encoding," *Energy*, vol. 328, p. 136572, Aug. 2025, doi: 10.1016/j.energy.2025.136572.

[111] M. M. Camboim *et al.*, "State of health estimation of second-life batteries through electrochemical impedance spectroscopy and dimensionality reduction," *J. Energy Storage*, vol. 78, p. 110063, Feb. 2024, doi: 10.1016/j.est.2023.110063.

[112] Y. Jiang, J. Jiang, C. Zhang, W. Zhang, Y. Gao, and N. Li, "State of health estimation of second-life LiFePO4 batteries for energy storage applications," *J. Clean. Prod.*, vol. 205, pp. 754–762, Dec. 2018, doi: 10.1016/j.jclepro.2018.09.149.

[113] F. Luo, H. Huang, L. Ni, and T. Li, "Rapid prediction of the state of health of retired power batteries based on electrochemical impedance spectroscopy," *J. Energy Storage*, vol. 41, p. 102866, Sep. 2021, doi: 10.1016/j.est.2021.102866.

[114] W. Zhang, T. Li, W. Wu, N. Ouyang, and G. Huang, "Data-driven state of health estimation in retired battery based on low and medium-frequency electrochemical impedance spectroscopy," *Measurement*, vol. 211, p. 112597, Apr. 2023, doi: 10.1016/j.measurement.2023.112597.

[115] M. Faraji-Niri, M. Rashid, J. Sansom, M. Sheikh, D. Widanage, and J. Marco, "Accelerated state of health estimation of second life lithium-ion batteries via electrochemical impedance spectroscopy tests and machine learning techniques," *J. Energy Storage*, vol. 58, p. 106295, Feb. 2023, doi: 10.1016/j.est.2022.106295.

[116] Y. Wang, H. Huang, H. Wang, and X. Wu, "A fast estimation method for state-of-health of retired batteries based on health features," *J. Energy Storage*, vol. 72, p. 108677, Nov. 2023, doi: 10.1016/j.est.2023.108677.

[117] E. Braco, I. San Martín, P. Sanchis, and A. Ursúa, "Fast capacity and internal resistance estimation method for second-life batteries from electric vehicles," *Appl. Energy*, vol. 329, p. 120235, Jan. 2023, doi: 10.1016/j.apenergy.2022.120235.

[118] M. Cheng, X. Zhang, A. Ran, G. Wei, and H. Sun, "Optimal dispatch approach for second-life batteries considering degradation with online SoH estimation," *Renewable and Sustainable Energy Reviews*, vol. 173, p. 113053, Mar. 2023, doi: 10.1016/j.rser.2022.113053.

[119] X. Cui, M. A. Khan, G. Pozzato, S. Singh, R. Sharma, and S. Onori, "Taking second-life batteries from exhausted to empowered using experiments, data analysis, and health estimation," *Cell Rep. Phys. Sci.*, vol. 5, no. 5, p. 101941, May 2024, doi: 10.1016/j.xcrp.2024.101941.

[120] S. Vignesh, H. S. Che, J. Selvaraj, and K. S. Tey, "State of health indicators for second life battery through non-destructive test approaches from repurposer perspective," *J. Energy Storage*, vol. 89, p. 111656, Jun. 2024, doi: 10.1016/j.est.2024.111656.

[121] S. M. Jameel, J. M. Altmemi, A. A. Oglah, M. A. Abbas, and A. H. Sabry, "Predicting batteries second-life state-of-health with first-life data and on-board voltage measurements using

support vector regression," *J. Energy Storage*, vol. 104, p. 114554, Dec. 2024, doi: 10.1016/j.est.2024.114554.

[122] S. Khaleghi *et al.*, "Developing an online data-driven approach for prognostics and health management of lithium-ion batteries," *Appl. Energy*, vol. 308, p. 118348, Feb. 2022, doi: 10.1016/j.apenergy.2021.118348.

[123] T. Montes, M. Etxandi-Santolaya, L. Canals Casals, and J. Eichman, "Multi-feature data-driven model for fast SOH assessment of batteries after their first life," *Math. Comput. Simul.*, vol. 238, pp. 150–162, Dec. 2025, doi: 10.1016/j.matcom.2025.05.008.

[124] G. Ma *et al.*, "A two-stage integrated method for early prediction of remaining useful life of lithium-ion batteries," *Knowl. Based. Syst.*, vol. 259, p. 110012, Jan. 2023, doi: 10.1016/j.knosys.2022.110012.

[125] Y. Zhang *et al.*, "Joint nonlinear-drift-driven Wiener process-Markov chain degradation switching model for adaptive online predicting lithium-ion battery remaining useful life," *Appl. Energy*, vol. 341, p. 121043, Jul. 2023, doi: 10.1016/j.apenergy.2023.121043.

[126] R. Wang, M. Zhu, X. Zhang, and H. Pham, "Lithium-ion battery remaining useful life prediction using a two-phase degradation model with a dynamic change point," *J. Energy Storage*, vol. 59, p. 106457, Mar. 2023, doi: 10.1016/j.est.2022.106457.

[127] W. Zhu *et al.*, "A mechanistic calendar aging model of lithium-ion battery considering solid electrolyte interface growth," *Int. J. Energy Res.*, vol. 46, no. 11, pp. 15521–15534, Sep. 2022, doi: 10.1002/er.8249.

[128] F. Pistorio, D. Clerici, F. Mocera, and A. Somà, "Review on the numerical modeling of fracture in active materials for lithium ion batteries," *J. Power Sources*, vol. 566, p. 232875, May 2023, doi: 10.1016/j.jpowsour.2023.232875.

[129] J. Shi, A. Rivera, and D. Wu, "Battery health management using physics-informed machine learning: Online degradation modeling and remaining useful life prediction," *Mech. Syst. Signal Process.*, vol. 179, p. 109347, Nov. 2022, doi: 10.1016/j.ymssp.2022.109347.

[130] M. El-Dalahmeh, M. Al-Greer, M. El-Dalahmeh, and I. Bashir, "Physics-based model informed smooth particle filter for remaining useful life prediction of lithium-ion battery," *Measurement*, vol. 214, p. 112838, Jun. 2023, doi: 10.1016/j.measurement.2023.112838.

[131] Bonnen Battery, "How Depth of Discharge (DOD) Affects EV Lithium Battery Lifespan." Accessed: May 25, 2026. [Online]. Available: https://www.bonnenbatteries.com/how-depth-of-discharge-dod-affects-ev-lithium-battery-lifespan/

[132] Danish Kapoor, "Understanding DC Fast Charging: Everything You Need to Know," EVSE. Accessed: Nov. 24, 2025. [Online]. Available: https://evse.com.au/blog/understanding-dc-fast-charging-everything-you-need-to-know/

[133] A. R. El Khatib, G. Hoblos, K. Langueh, and E. Duviella, "From First Life to Second Life: Advances and Research Gaps in Prognosis Techniques for Lithium-Ion Batteries," *Applied Sciences*, vol. 15, no. 22, p. 12171, Nov. 2025, doi: 10.3390/app152212171.

[134] S. Ansari, A. Ayob, M. S. Hossain Lipu, A. Hussain, and M. H. M. Saad, "Multi-Channel Profile Based Artificial Neural Network Approach for Remaining Useful Life Prediction of Electric Vehicle Lithium-Ion Batteries," *Energies (Basel).*, vol. 14, no. 22, p. 7521, Nov. 2021, doi: 10.3390/en14227521.

[135] S. Sohn, H.-E. Byun, and J. H. Lee, "CNN-based Online Diagnosis of Knee-point in Li-ion Battery Capacity Fade Curve," *IFAC-PapersOnLine*, vol. 55, no. 7, pp. 181–185, 2022, doi: 10.1016/j.ifacol.2022.07.441.

[136] W. Gao *et al.*, "Comprehensive study of the aging knee and second-life potential of the Nissan Leaf e+ batteries," *J. Power Sources*, vol. 613, p. 234884, Sep. 2024, doi: 10.1016/j.jpowsour.2024.234884.

[137] E. Braco, I. San Martín, A. Berrueta, P. Sanchis, and A. Ursúa, "Experimental assessment of cycling ageing of lithium-ion second-life batteries from electric vehicles," *J. Energy Storage*, vol. 32, p. 101695, Dec. 2020, doi: 10.1016/j.est.2020.101695.

[138] Z. Lv *et al.*, "Empirical model, capacity recovery-identification correction and machine learning co-driven Li-ion battery remaining useful life prediction," *J. Energy Storage*, vol. 103, p. 114274, Dec. 2024, doi: 10.1016/j.est.2024.114274.

[139] M. S. Hosen, J. Jaguemont, J. Van Mierlo, and M. Berecibar, "Battery lifetime prediction and performance assessment of different modeling approaches," *iScience*, vol. 24, no. 2, p. 102060, Feb. 2021, doi: 10.1016/j.isci.2021.102060.

[140] Z. Xue, Y. Zhang, C. Cheng, and G. Ma, "Remaining useful life prediction of lithium-ion batteries with adaptive unscented kalman filter and optimized support vector regression," *Neurocomputing*, vol. 376, pp. 95–102, Feb. 2020, doi: 10.1016/j.neucom.2019.09.074.

[141] S. V. Ganesh and M. D'Arpino, "Critical Comparison of Li-Ion Aging Models for Second Life Battery Applications," *Energies (Basel).*, vol. 16, no. 7, p. 3023, Mar. 2023, doi: 10.3390/en16073023.

[142] L. Marthaler, P. Grudzien, M. Buysse, M. Ierides, and A. Mccready, "Key technical, policy and market developments influencing the electric vehicle battery landscape," Bax and Company. Accessed: Jul. 25, 2025. [Online]. Available: https://projectcobra.eu/wp-content/uploads/2022/12/Market-Intelligence-Report-December-2022.pdf

[143] RWE generation SE, "Second life for EV batteries: RWE and Audi create novel energy storage system in Herdecke." Accessed: May 22, 2025. [Online]. Available: https://www.rwe.com/en/press/rwe-generation/2021-12-28-second-life-for-ev-batteries/

[144] Fortum, "Fortum installs innovative battery solution at Landafors hydropower plant in Sweden." Accessed: Nov. 23, 2025. [Online]. Available: https://www.fortum.com/media/2021/04/fortum-installs-innovative-battery-solution-landafors-hydropower-plant-sweden

[145] "Enel launches innovative,'Second Life' storage system for used electric car batteries in Melilla, Spain," Enel Group. Accessed: Jan. 20, 2026. [Online]. Available: https://www.enel.com/media/explore/search-press-releases/press/2022/03/enel-launches-innovative-second-life-storage-system-for-used-electric-car-batteries-in-melilla-spain-

[146] Jungheinrich, "JT Energy Systems builds 25 MW battery storage system." Accessed: Nov. 24, 2025. [Online]. Available: https://www.jungheinrich.com/en/newsroom/jt-energy-systems-builds-25-mw-battery- storage-system-1097964

[147] Fortum, "Second life batteries, expertise and a can-do attitude – innovation boosts hydropower plant." Accessed: Nov. 25, 2025. [Online]. Available: https://www.fortum.com/about-us/our-company/strategy/decarbonisation-joint-effort/second-life-batteries-hydropower-plant

[148] Ö. Özcan, A. C. Duman, Ö. Gönül, and Ö. Güler, "Techno-economic analysis of grid-connected PV and second-life battery systems for net-zero energy houses," *Journal of Building Engineering*, vol. 89, p. 109324, Jul. 2024, doi: 10.1016/j.jobe.2024.109324.

[149] N. Kebir *et al.*, "Second-life battery systems for affordable energy access in Kenyan primary schools," *Sci. Rep.*, vol. 13, no. 1, p. 1374, Jan. 2023, doi: 10.1038/s41598-023-28377-7.

[150] A. Hassan *et al.*, "Second-Life Batteries: A Review on Power Grid Applications, Degradation Mechanisms, and Power Electronics Interface Architectures," *Batteries*, vol. 9, no. 12, p. 571, Nov. 2023, doi: 10.3390/batteries9120571.

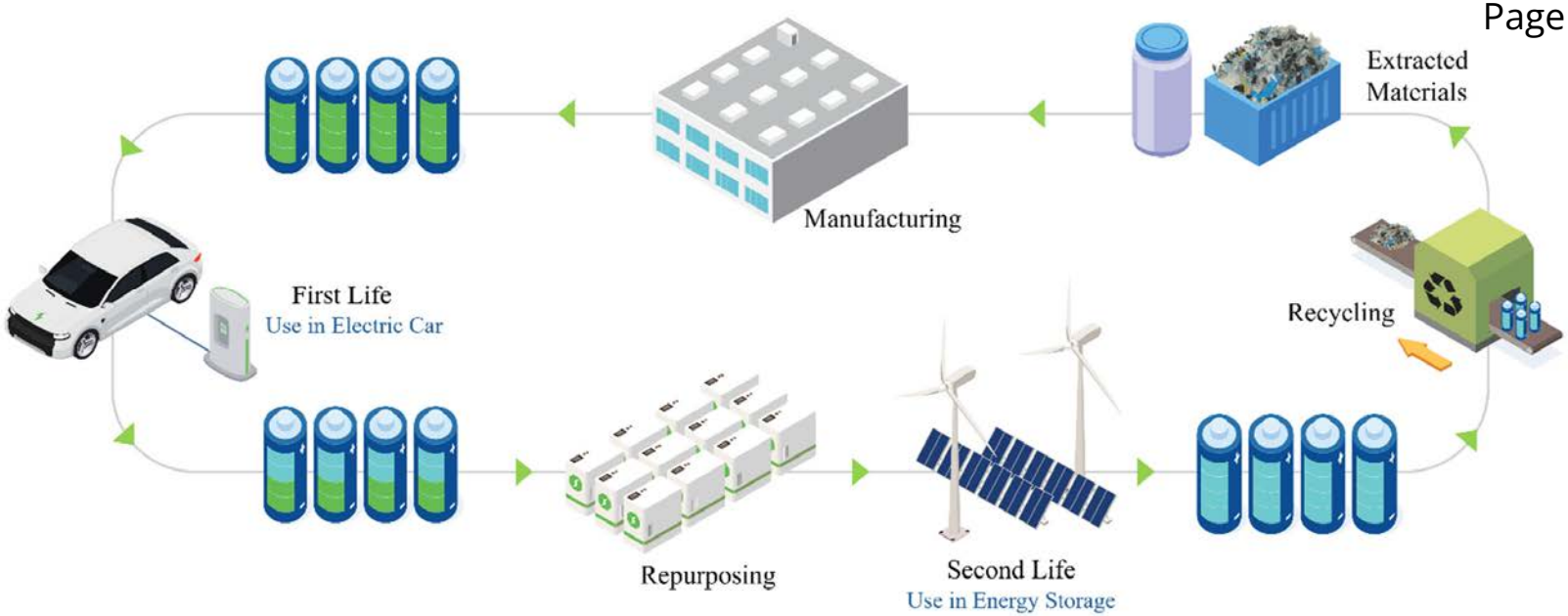
[151] J. Thakur, C. Martins Leite de Almeida, and A. G. Baskar, "Electric vehicle batteries for a circular economy: Second life batteries as residential stationary storage," *J. Clean. Prod.*, vol. 375, p. 134066, Nov. 2022, doi: 10.1016/j.jclepro.2022.134066.

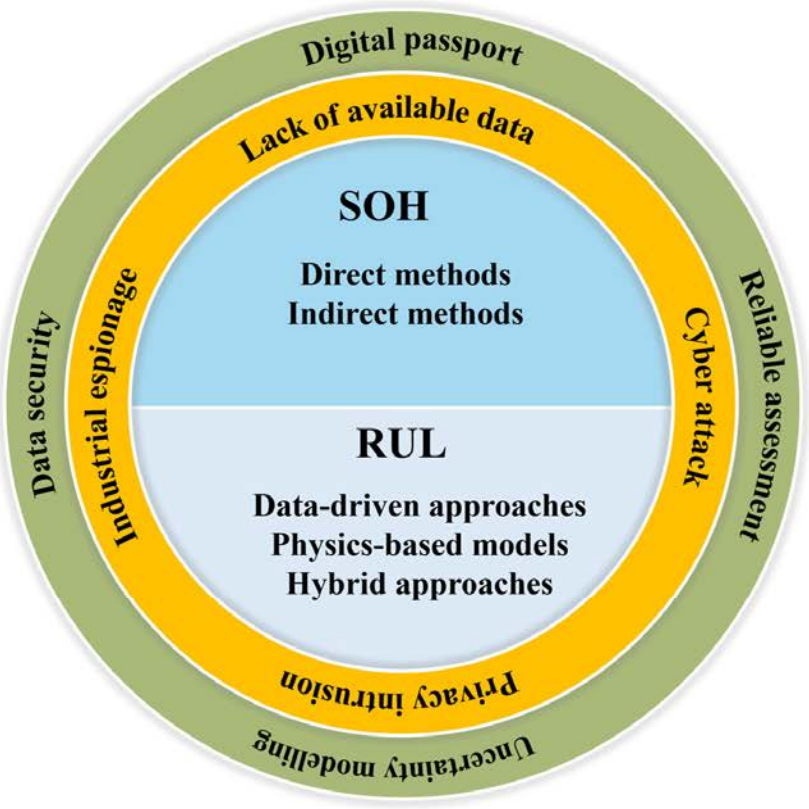
[152] K. Neigum and Z. Wang, "Technology, economic, and environmental analysis of second-life batteries as stationary energy storage: A review," *J. Energy Storage*, vol. 103, p. 114393, Dec. 2024, doi: 10.1016/j.est.2024.114393.

[153] Y. Zhao *et al.*, "A Review on Battery Market Trends, Second-Life Reuse, and Recycling," *Sustainable Chemistry*, vol. 2, no. 1, pp. 167–205, Mar. 2021, doi: 10.3390/suschem2010011.

[154] C. Antônio Rufino Júnior, E. R. Sanseverino, P. Gallo, D. Koch, H.-G. Schweiger, and H. Zanin, "Blockchain review for battery supply chain monitoring and battery trading," *Renewable and Sustainable Energy Reviews*, vol. 157, p. 112078, Apr. 2022, doi: 10.1016/j.rser.2022.112078.

[155] H. Song, H. Chen, Y. Wang, and X.-E. Sun, "An Overview About Second-Life Battery Utilization for Energy Storage: Key Challenges and Solutions," *Energies (Basel).*, vol. 17, no. 23, p. 6163, Dec. 2024, doi: 10.3390/en17236163.


Extracted
Materials
Manufacturing
First Life
Use in Electric Car
Recycling
Repurposing
Second Life
Use in Energy Storage


Digital passport
Lack of available data
SOH
Direct methods
Indirect methods
RUL
Data-driven approaches
Physics-based models
Hybrid approaches
Data security
Industrial espionage
Cyber attack
Reliable assessment
Privacy intrusion
Uncertainty modelling